Kinematics-1

(motion in 1D)

* rest and motion are relative but not absolute.

Frame of reference:- (→ ~~either rest or~~)

Inertial frame of refuence:→ either rest or in uniform motion.

Non-inertial frame of refuence:→ accelerate

* Co-ordinate axes system is called frame of refuence.

Distance:-

* Actual path length travelled by the body b/w initial & final positions.

* It is a scalar & is always +ve.

* Units:- m, cm

* In integration form,

$$\text{distance} = \int \sqrt{(dx)^2 + (dy)^2}$$

$$D = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$ — applicable only when $y = f(x)$

$$D = \int \sqrt{1 + (f'(x))^2}\, dx$$

1 Km = 1000m

1 mile = 1.6 Km

$1 A^\circ = 10^{-10} m$

$1 \mu m = 10^{-6} m$

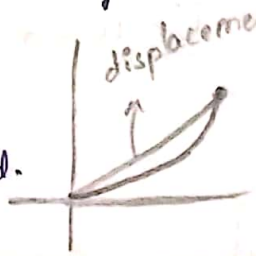
Displacement:-

* Shortest path length b/w initial & final positions.

* It is a vector and it's direction is always from initial position to final position.

* It doesnot depends on path followed.

* It's Units are same as distance.



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* The body which is moving in a straight line path is said to be 1D motion.

* In straight line path, distance & displacement are always equal.

* In general,

$$|\text{distance}| \geq |\text{displacement}|$$

or

$$0 \leq |\text{displacement}| \leq |\text{distance}|$$

* Usually, $\overline{\Delta x}$ = displacement

$$|\overline{\Delta x}| \neq \text{distance}.$$

* displacement is zero when the body travelled from initial position to final position & come back to initial position.

Average Velocity & Average Speed:-

① Average speed :-

$$\langle v \rangle = \frac{\text{Total distance}}{\text{Total time}}$$

* Avg speed is a scalar quantity.

* Units → m/s or km/hr

$$\boxed{11\text{km/hr} = \frac{5}{18}\text{m/s}}$$

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② Average Velocity :-

$$\langle V \rangle = \frac{\text{net displacement}}{\text{Total time}} = \frac{\overline{\Delta x}}{\Delta t}$$

$$\boxed{\frac{\overline{\Delta x}}{\Delta t} = \frac{\bar{x}_f - \bar{x}_i}{t_2 - t_1}}$$

* It is a vector quantity.
* It's direction is along $\overline{\Delta x}$ (net displacement).
* It may be positive, negative or zero.
* $|\overline{\Delta x}| \leq D$ (total distance)
* $\frac{|\overline{\Delta x}|}{\Delta t} \leq \frac{D}{\Delta t}$

$|\text{avg velocity}| \leq \text{avg speed}$.

* both are equal when the body is moving in straight line with out turnings.

Instantaneous speed (speed) :-

Speed of a body at a particular instant of time.

$$\boxed{v = \lim_{\Delta t \to 0} \frac{\Delta D}{\Delta t} = \frac{dD \to \text{distance}}{dt}}$$

* It is a scalar. It is always +ve.

Instantaneous velocity (velocity) :-

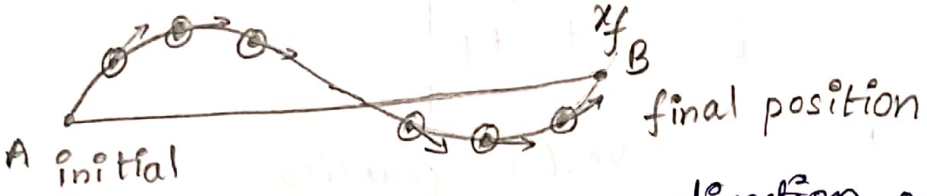
Velocity of a body at a particular instant of time.

$$\boxed{\bar{v} = \lim_{\Delta t \to 0} \frac{\Delta \bar{s}}{\Delta t} = \frac{d\bar{s}}{dt}}$$

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* It is a vector. It's direction is always "along" the direction of motion.

$$\langle v \rangle = \frac{\Delta \bar{x}}{\Delta t} = \frac{\bar{x}_f - \bar{x}_i}{t_f - t_i}$$



direction of motion = direction of velocity at the point.

* velocity at a point may be +ve, -ve or zero.

* |velocity| = speed.

Uniform Motion :-

* Velocity = const i.e, both magnitude and direction remains constant.

* In uniform motion, avg velocity & instantaneous velocity both are same.

* $\bar{v}$ — const $\Rightarrow |\bar{v}| = v = \text{const}$.

* In uniform motion, $v = \frac{d}{t}$

$$\Rightarrow \boxed{d = vt} \text{ (or) } \boxed{\bar{s} = \bar{v}t}$$

** $\boxed{\bar{x}_f - \bar{x}_i = vt}$

Applications :-

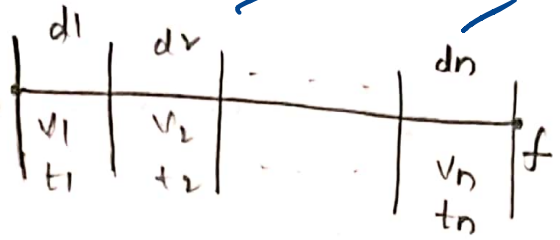
① If journey is divided into n-equal path

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lengths :-

$d_1 + d_2 + \cdots + d_n = d$

$nd_1 = d \Rightarrow \boxed{d_1 = d/n}$

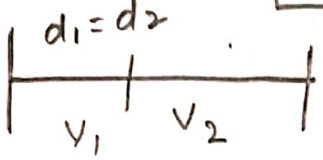


Average speed

$$V = \frac{\text{Total distance}}{\text{Total time}} = \frac{d_1 + d_2 + \cdots + d_n}{t_1 + t_2 + \cdots + t_n}$$

$$V = \frac{d}{\frac{d}{nv_1} + \frac{d}{nv_2} + \cdots + \frac{d}{nv_n}}$$

$$\boxed{\frac{n}{V} = \frac{1}{v_1} + \frac{1}{v_2} + \cdots + \frac{1}{v_n}} \rightarrow H.P$$

→ 

$$\frac{2}{V} = \frac{1}{v_1} + \frac{1}{v_2}$$

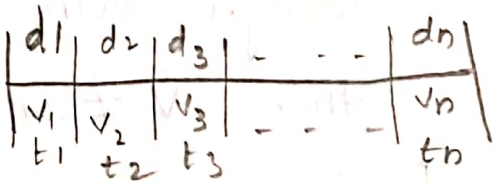
$$\boxed{\frac{2v_1v_1}{v_1 + v_2} = V}^{*} \rightarrow \text{Harmonic mean}$$

② If journey is divided into n-equal time intervals

$t = t_1 + t_2 + \cdots + t_n$

$nt_1 = t$

$\boxed{t_1 = t/n}$

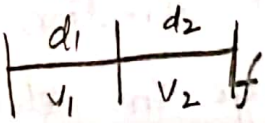


$d_1 = v_1t_1 \; ; \; d_2 = v_2t_2 \cdots d_n = v_nt_n$

$$V = \frac{d_1 + d_2 + \cdots + d_n}{t_1 + t_2 + \cdots + t_n} = \frac{v_1t_1 + v_2t_2 + \cdots + v_nt_n}{t}$$

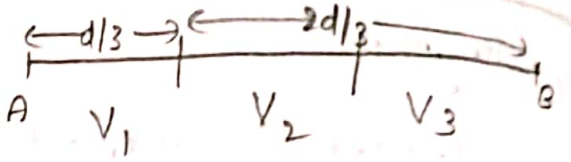
$$\boxed{V = \frac{v_1 + v_2 + v_3 + v_4 + \cdots + v_n}{n}} \rightarrow A.P$$

→ $\boxed{V = \frac{v_1 + v_2}{2}}$ → arithmetic mean



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→



$$\frac{d}{3} = v_1 t_1$$

$$\frac{2d}{3} = v_2 t_2 + v_3 t_2 \Rightarrow t_2 = \frac{2d}{3(v_2+v_3)}$$

$$\therefore v = \frac{d}{t_1+t_2+t_2} = \frac{d}{t_1+2t_2}$$

$$v = \frac{d}{\frac{d}{3v_1} + \frac{4d}{3(v_2+v_3)}}$$

$$\boxed{v = \frac{3v_1(v_2+v_3)}{v_2+v_3+4v_1}}$$

Non-Uniform Motion :-

* $\bar{v}$ is not constant.
* only magnitude changes, only direction changes, both magnitude and direction change then $\bar{v} \neq$ const.
* Here body possess acceleration.

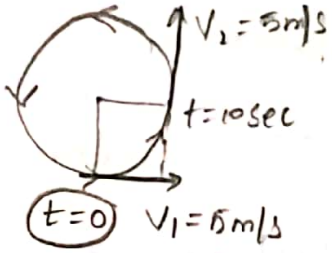
Average acceleration :-

$$\langle \bar{a} \rangle = \frac{\Delta \bar{v}}{\Delta t} = \frac{\bar{v}_f - \bar{v}_i}{t_f - t_i}$$

* it is a vector quantity.
* Its direction is always in the direction of change in velocity.

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$\bar{v}_1 = 5\hat{\uparrow}$, $\bar{v}_2 = 5\hat{\uparrow}$



$$\langle \bar{a} \rangle = \frac{\bar{v}_2 - \bar{v}_1}{10}$$

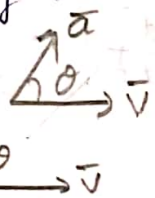
$$\langle \bar{a} \rangle = \frac{5\hat{\uparrow} - 5\hat{\uparrow}}{10}$$

$$\boxed{\langle \bar{a} \rangle = \frac{1}{2}(\hat{\uparrow} - \hat{\uparrow})}$$

Instantaneous acceleration :-

$$\bar{a} = \lim_{\Delta t \to 0} \frac{\Delta \bar{v}}{\Delta t} = \frac{d\bar{v}}{dt}$$

* $\bar{a}$ direction is not always along $\bar{v}$.
* If $0° \leq \theta < 90°$ then speed of particle increases.



* If $90 < \theta \leq 180°$ then speed of particle decreases.

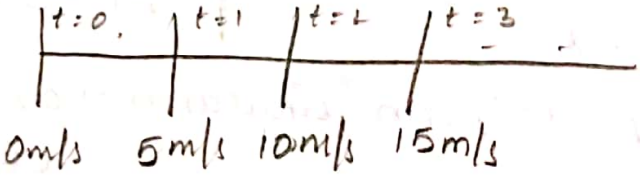


* If $\theta = 90°$ then speed of particle remains constant.



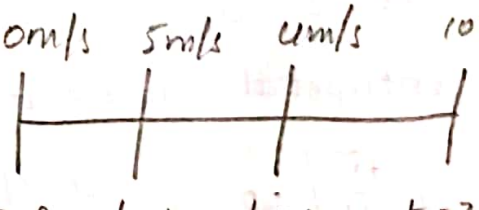
Uniform acceleration :-

* The change in velocity is equal in equal intervals of time is called uniform acceleration.
* $\bar{a}$ = constant.



Uniform or variable acceleration :-

* $\bar{a}$ → not constant
* Velocity is varying in equal intervals of time.



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Points:-

1. $\bar{v}$ = velocity.
2. $|\bar{v}|$ = speed
3. $\bar{a} = \frac{d\bar{v}}{dt}$ = Total acceleration. (net acceleration)
4. $|\bar{a}| = \left|\frac{d\bar{v}}{dt}\right|$ = magnitude of total acceleration.
5. $a_t = \frac{d|\bar{v}|}{dt}$ = rate of change of speed called tangential acceleration, one of the component of total acceleration.
6. $\left|\frac{d\bar{v}}{dt}\right| = \frac{d}{dt}|\bar{v}|$; we have $\bar{v} = |\bar{v}| \cdot \hat{v}$

$$\frac{d\bar{v}}{dt} = \frac{d}{dt}|\bar{v}| \cdot \hat{v}$$

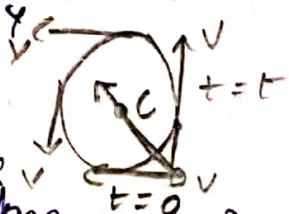
$$\frac{d\bar{v}}{dt} = \hat{v} \cdot \frac{d|\bar{v}|}{dt}$$

for a translational motion with $\bar{v} \leftarrow f(t)$

$$\boxed{\left|\frac{d\bar{v}}{dt}\right| = \frac{d|\bar{v}|}{dt}}$$

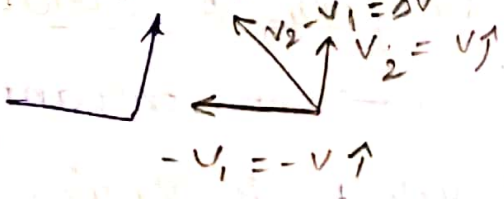
7. $\frac{d|\bar{v}|}{dt} = 0$ then $\left|\frac{d\bar{v}}{dt}\right| = 0$ is not always true.

Eg: uniform circular motion.



(velocity is tangential along centre)

then $\Delta\bar{v} = \bar{v}_2 - \bar{v}_1 = \bar{v}_2 + (-\bar{v}_1)$



it is also called as centripetal acceleration

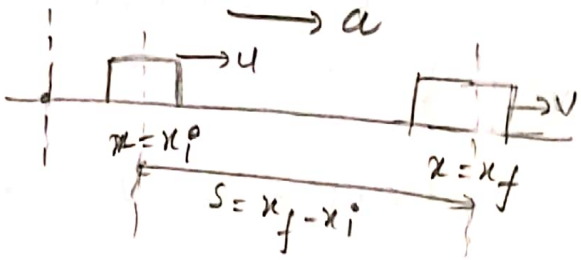
8. If $\left|\frac{d\bar{v}}{dt}\right| = 0$ the $\frac{d|\bar{v}|}{dt} = 0$ is always true.

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Equations of motion for uniformly accelerated motions:-

1. $\bar{v} = \bar{u} + \bar{a}t$

2. $\bar{s} = \bar{x}_f - \bar{x}_i = \bar{u}t + \frac{1}{2}\bar{a}t^2$



3. $v^2 - u^2 = 2as \Rightarrow \bar{v}\cdot\bar{v} - \bar{u}\cdot\bar{u} = 2\bar{a}\cdot\bar{s}$

4. $S_n = u + \frac{a}{2}(2n-1)$ → $n \to n^{th}$ second (time)

5. $s = \left(\frac{\bar{u}+\bar{v}}{2}\right)t$

$S_n = u + \frac{a}{2}(2n-1)$

* de-acceleration is also called negative acceleration.

→ given $S_n = u + 0.2n$. Find u & a.

$$S_n = u + 0.2n + \frac{0.1}{2} - \frac{0.2}{2}$$

$$S_n = 4.1 + 0.2\left(n - \frac{1}{2}\right)$$

$$S_n = 4.1 + \frac{0.2}{2}(2n-1)$$

$\Rightarrow u = 4.1$, $a = 0.2$

→ given $S_n = 7 + 0.5n$. Find u & a.

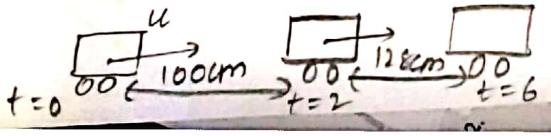
$$S_n = 7 + 0.5n + \frac{0.5}{2} - \frac{0.5}{2}$$

$$= \frac{14.5}{2} + 0.5\left(n - \frac{1}{2}\right)$$

$\Rightarrow a = 0.5$, $u = 7.25$ ll

→ A body covers 100 cm in first two seconds and 128 cm in next four seconds. Find velocity of the body at the end of 8th sec.

WKT, $s = ut + \frac{1}{2}at^2$.



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$$100 = u(2) + \frac{1}{2}a(2)^2$$

$$100 = 2u + 2a$$

$$\boxed{u + a = 50} \quad -①$$

$$100 + 128 = u(6) + \frac{1}{2}a(6)^2$$

$$228 = 6u + 18a$$

$$114 = 3u + 9a$$

$$\boxed{38 = u + 3a} \quad -②$$

② - ① $\Rightarrow$ $2a = 38 - 50$

$$2a = -12$$

$$\boxed{a = -6 \text{ cm/s}^2}$$

$\Rightarrow$ $u = 50 + 6$

$$\boxed{u = 56 \text{ cm/s}}$$

$$V = u + at$$

$$= 56 - 6(8)$$

$$= 56 - 48$$

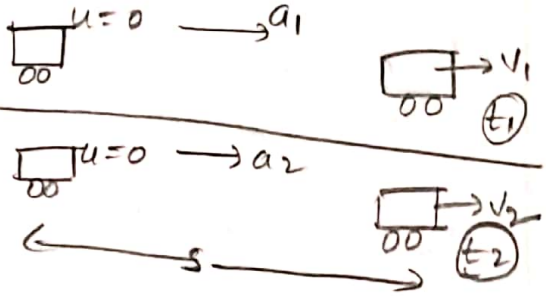
$$\boxed{V = 8 \text{ cm/s}}$$

→ In a car race, car A takes time 't' less than car B and passes the finishing point with a velocity 'v' more than the velocity with which car B passes the point. Assuming the cars start from rest and travels with const acceleration

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Given $t_1 = t_2 - t \Rightarrow t = t_2 - t_1$ ——①

& $V_1 = V_2 + V \Rightarrow V = V_1 - V_2$ ——②



now, $\frac{V}{t} = \frac{V_1 - V_2}{t_2 - t_1}$

WKT, $v^2 - u^2 = 2as$

$V_1^2 - 0^2 = 2a_1 s$ $\quad$ $V_2^2 - 0^2 = 2a_2 s$

$V_1 = \sqrt{2a_1 s}$ $\quad$ $V_2 = \sqrt{2a_2 s}$

$s = ut + \frac{1}{2}at^2$

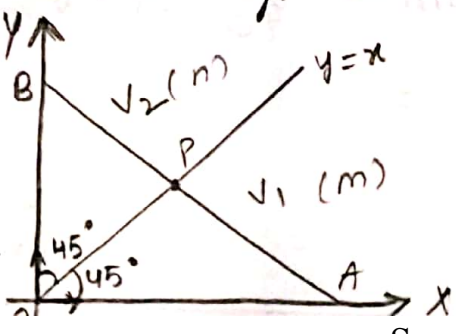
$s = \frac{1}{2}a_1 t_1^2 \Rightarrow t_1 = \sqrt{\frac{2s}{a_1}}$

$s = \frac{1}{2}a_2 t_2^2 \Rightarrow t_2 = \sqrt{\frac{2s}{a_2}}$

$$\frac{V}{t} = \frac{\sqrt{2a_1 s} - \sqrt{2a_2 s}}{\sqrt{\frac{2s}{a_1}} - \sqrt{\frac{2s}{a_2}}} = \frac{\sqrt{a_1} - \sqrt{a_2}}{\frac{1}{\sqrt{a_2}} - \frac{1}{\sqrt{a_1}}}$$

$$\frac{V}{t} = \frac{(\sqrt{a_1} - \sqrt{a_2})\sqrt{a_1 a_2}}{(\sqrt{a_1} - \sqrt{a_2})} = \sqrt{a_1 a_2}$$

→ Three particles starts from the origin at the same line, one with a velocity V_1 along X-axis, the second along Y-axis with a velocity V_2 and the third along (x=y) line. The velocity of third, so that the three may ~~touch~~ always lie on the same line is.

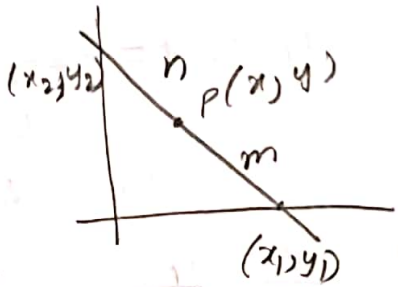


Similar Δles,

$$\frac{PA}{OA} = \frac{PB}{OB} \Rightarrow \frac{PA}{PB} = \frac{OA}{OB}$$

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$$\frac{PA}{PB} = \frac{v_1}{v_2}$$



$$v_x = \frac{v_1(0) + v_2(v_1)}{v_1 + v_2} = \frac{v_1 v_2}{v_1 + v_2} \qquad x = \frac{mx_2 + nx_1}{m+n}$$

$$v_y = \frac{v_1 v_2 + v_2(0)}{v_1 + v_2} = \frac{v_1 v_2}{v_1 + v_2} \qquad y = \frac{my_2 + ny_1}{m+n}$$

$$\bar{v} = v_x \uparrow + v_y \uparrow$$

$$|\bar{v}| = \frac{\sqrt{2}\, v_1 v_2}{v_1 + v_2}$$

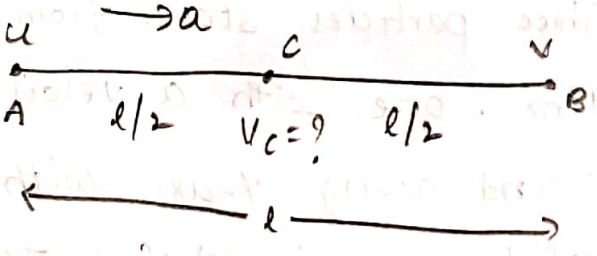
Like train Ques

→ A point moving with const acceleration from A to B in a straight line AB has velocities u and v at A and B respectively.

(a) Find the velocity at C (the mid point of A & B)

(b) Find the ratio $\frac{v}{u}$, if the time taken from A to C is twice the time to go from C to B.

a)



for AC, $v_c^2 - u^2 = 2al/2$ — ①

for CB, $v^2 - v_c^2 = 2al/2$ — ②

From ① & ② ⇒ $2v_c^2 = \frac{v^2+u^2}{1}$

① = ②

$$v_c^2 = \frac{v^2+u^2}{2}$$

$$v_c = \sqrt{\frac{v^2+u^2}{2}}$$

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b) $S = \left(\frac{V+u}{2}\right) t$

$\frac{d}{2} = \left(\frac{u+V_{ac}}{2}\right) t_{AC}$

$\frac{d}{2} = \left(\frac{V_c+V}{2}\right) t_{BC}$

given $t_{AC} = 2 t_{BC}$

$\frac{d}{x} \times \left(\frac{x}{u+V_{ac}}\right) = 2 \times \frac{t}{2} \left(\frac{2}{V_c+V}\right)$

$V_c + V = 2u + 2V_{ac}$

$V - 2u = V_c$

$V - 2u = \sqrt{\frac{V^2+u^2}{2}}$

$(V-2u)^2 = \frac{V^2+u^2}{2}$

$\left(\frac{V}{u} - \frac{2u}{u}\right)^2 = \frac{\left(\frac{V}{u}\right)^2 + 1}{2}$

let $\frac{V}{u} = p$ $\quad (p-2)^2 = \frac{p^2+1}{2}$

$p^2 - 4p + 4 = \frac{p^2+1}{2}$

$2p^2 - 8p + 8 = p^2 + 1$

$p^2 - 8p + 7 = 0$

$p = \frac{8 \pm \sqrt{64-28}}{2} = \frac{8 \pm 6}{2} \Rightarrow \frac{8+6}{2}, \frac{8-6}{2}$

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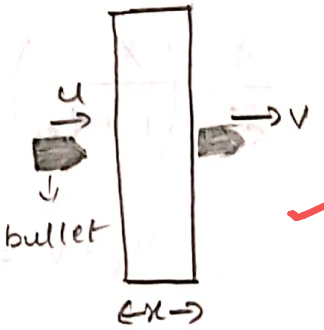
$p = 7, 1$

$v \neq u$

$p = 7 \Rightarrow \boxed{\frac{v}{u} = 7}$

→ If the bullet loses $\frac{1}{n}$ th of initial velocity after passing through one plank, Find number of such planks required to stop the bullet?

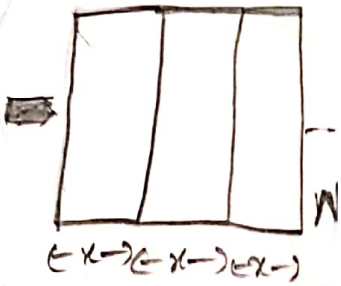
$$v = u - \frac{u}{n} = u\left(1 - \frac{1}{n}\right)$$



$$v^2 - u^2 = 2as$$

$$u^2\left(1 - \frac{1}{n}\right)^2 - u^2 = 2ax \quad -(1)$$

$$0^2 - u^2 = 2aNx \quad -(2)$$



$$\frac{(1)}{(2)} \Rightarrow \frac{u^2\left(\left(1 - \frac{1}{n}\right)^2 - 1\right)}{-u^2} = \frac{2ax}{2aNx}$$

$$1/N = 1 - \left(1 - \frac{1}{n}\right)^2$$

$$1/N = 1 - 1 - \frac{1}{n^2} + \frac{2}{n}$$

$$1/N = \frac{2}{n} - \frac{1}{n^2}$$

$$1/N = \frac{2n - 1}{n^2}$$

$$\boxed{N = \frac{n^2}{2n-1}}$$

$v^2 - u^2 = -2a[nx]$

$n = -$

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$$v = u - \frac{u}{n} = u\left(1 - \frac{1}{n}\right)$$

$$v^2 - u^2 = 2a(x)$$

$$u^2\left(1 - \frac{1}{n}\right)^2 - u^2 = 2a(x)$$

$$\frac{u^2\left(1 - \frac{1}{n}\right)^2 - u^2}{2x} = a$$

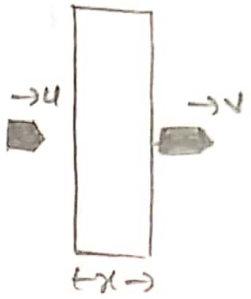
$$v^2 - u^2 = 2a(\eta(x))$$

$$u^2 = x\left[\frac{u^2\left[\left(1 - \frac{1}{n}\right)^2 - 1\right]}{2x}\right].$$

$$n' = \frac{1}{1 + \frac{1}{n^2} - \frac{2}{n} - 1}$$

$$n = \frac{n^2}{1 - 2n}$$

If the velocity of bullet becomes $\frac{1}{n}$th of the initial velocity. Find no. of planks to stop the bullet?



$$v = \frac{u}{n}$$

$$v^2 - u^2 = 2as$$

$$\frac{u^2}{n^2} - u^2 = 2ax \quad -(1)$$

$$0 - u^2 = 2a(-x) \quad -(2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{u^2\left(\frac{1-n^2}{n^2}\right)}{-u^2} = \frac{2ax}{2a(-x)}$$

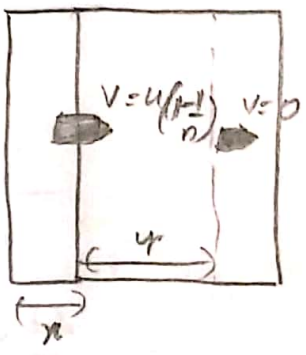
$$\frac{n^2-1}{n^2} = \frac{1}{N}$$

$$\boxed{N = \frac{n^2}{n^2-1}}$$

After penetrating a distance 'x' bullet becomes $\frac{1}{n}$th of initial velocity. Find further distance travelled by the bullet before coming to rest.

$$u^2\left(1-\frac{1}{n}\right)^2 - u^2 = 2ax \quad -(1)$$

$$0^2 - u^2 = 2a(x+y) \quad -(2)$$



$$\frac{(1)}{(2)} \quad \frac{u^2\left(\left(1-\frac{1}{n}\right)^2 - 1\right)}{-u^2} = \frac{x}{x+y}$$

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$$1-\left(1-\frac{1}{n}\right)^2 = \frac{x}{x+y}$$

$$1-1+\frac{1}{n^2}-\frac{2}{n} = \frac{x}{x+y}$$

$$\frac{x+y}{x} = \frac{1}{\frac{2n-1}{n^2}}$$

$$1+\frac{y}{x} = \frac{n^2}{2n-1}$$

$$\frac{y}{x} = \frac{n^2-2n+1}{2n-1}$$

$$\boxed{y = \frac{(n-1)^2}{2n-1}x}$$

→ If the velocity of bullet becomes $\frac{1}{n}$th of initial velocity after penetrating a distance x.

$$\left(\frac{u}{n}\right)^2 - u^2 = 2ax \quad -①$$

$$0^2 - u^2 = 2a(x+y) \quad -②$$

$$\frac{①}{②} \Rightarrow \frac{u^2\left(\frac{1-n^2}{n^2}\right)}{-u^2} = \frac{2ax}{2a(x+y)}$$

$$\frac{n^2-1}{n^2} = \frac{x}{x+y}$$

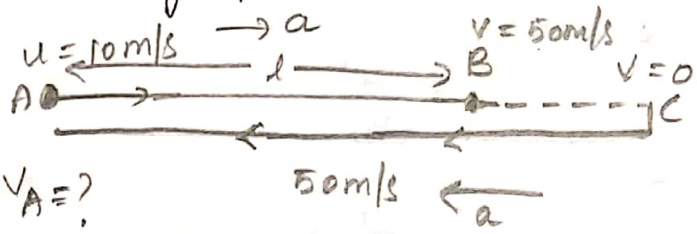
$$\frac{x+y}{x} = \frac{n^2}{n^2-1}$$

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$$\frac{y}{x} = \frac{n^2 - n^2 + 1}{n^2 - 1}$$

$$\boxed{y = \frac{x}{n^2 - 1}}$$

A body starts with an intial velocity of 10m/s and moving along a straight line path with constant accelesation. when the velocity of the body is 50m/s, the acceleration is reversed in direction. Find the velocity of the particle as it reaches the starting point.



$$V_B^2 - u^2 = 2al \quad -①$$

$$V_A^2 - V_B^2 = 2al \quad -②$$

$$V_B^2 - u^2 = V_A^2 - V_B^2$$

$$V_A^2 = 2V_B^2 - u^2$$

$$V_A^2 = 2(2500) - 100$$

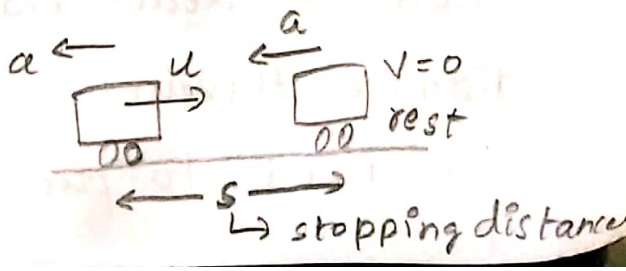
$$= 5000 - 100$$

$$V_A^2 = 4900$$

$$\boxed{V_A = 70\,m/s}$$

Same Vel.

Stoping distance:-



$$v^2 - u^2 = 2as$$

$$0^2 - u^2 = 2as$$

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$$S = \frac{-u^2}{2a}$$

$$S = \frac{u^2}{2(-a)}$$

$$\boxed{S = \frac{u^2}{2 \times \text{retardation}}}$$

Stopping distance

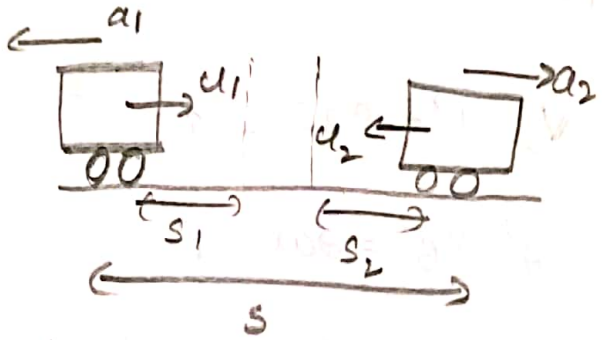
$S \propto u^2$ [if $-a$ = const]

$$\boxed{\frac{s_1}{s_2} = \left(\frac{u_1}{u_2}\right)^2}$$

*

$$S_1 = \frac{u^2}{2a_1}$$

$$S_2 = \frac{u^2}{2a_2}$$



If $s_1 + s_2 < S \longrightarrow$ no collision takes place.

$s_1 + s_2 > S \longrightarrow$ collision takes place.

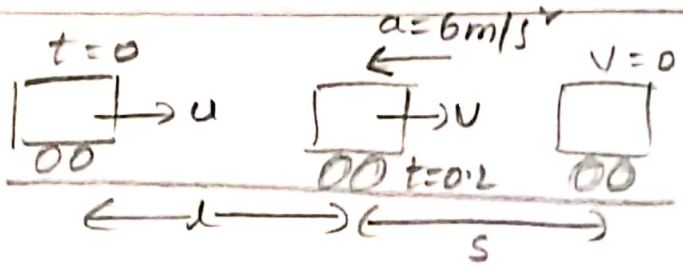
Reaction time :-

→ A driver takes 0.2sec to apply the breaks after he sees a need for it. This is called rxn time of the driver. If he is driving at a speed of 54 km/hr and the breaks causes a deacceleration of $6 m/s^2$. Find the distance travelled by the car after he sees the need to put the breaks.

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$l = u \times t = 15 \times 0.2$

$\boxed{l = 3m}$



$S = \frac{u^2}{2(-a)} = \frac{15 \times 15}{2 \times 6_2} = \frac{75}{4}$

$u = 54 km/hr$

$= 54 \times 5/18$

$\boxed{u = 15 m/s}$

$S = \frac{75}{4} m$

∴ Total distance $= 3 + \frac{75}{4} = \frac{12+75}{4} = \frac{87}{4}$

$= 21.75m$

Two Tourists A & B, they are at a distance of 40km from the camp. They want to go to the camp at a time. But they have only one cycle. Only one member can ride a cycle. person A rode the cycle for some distance and left the cycle there and started walking. person B walked to some distance to the cycle and started riding. They will go to the camp at a time. If the walking speed is 5 kmph & cycling speed is 15 kmph. Find

① Average speed of each person.

② Find the time that the cycle is left used?.

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Variable acceleration :-

* $a \neq$ const.
* $a = \frac{dv}{dt}$
* $a = \frac{d}{dt}\left(\frac{ds}{dt}\right) = \frac{d^2s}{dt^2}$
* $a = \frac{dv}{ds} \times \frac{ds}{dt} = v \cdot \frac{dv}{ds}$
* $\frac{da}{dt} \rightarrow$ jerk.

→ A bird flies for four seconds with a velocity $(v = t - 2)$ m/s in a straight line, where t is in seconds. Calculate

① displacement and ② distance travelled by the bird.

M-I:

1) displacement $(s) = \int_0^4 v \cdot dt = \int_0^4 (t-2)\, dt$

$$= \left[\frac{t^2}{2} - 2t\right]_0^4 = [8 - 8]$$

$$= 0$$

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2) distance, $v = 0$

$t - 2 = 0$

$t = 2$ sec.

$$\text{distance} = \int_0^2 |t-2| \cdot dt + \int_2^4 |t-2| \cdot dt$$

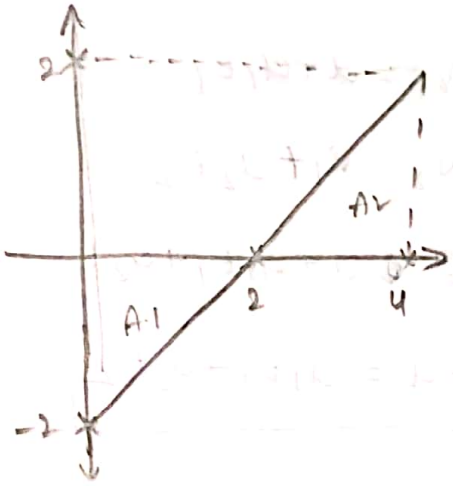
$$= \int_0^2 (2-t) \cdot dt + \int_2^4 (t-2) \cdot dt$$

$$= \left[2t - \frac{t^2}{2}\right]_0^2 + \left[\frac{t^2}{2} - 2t\right]_2^4$$

$$= 2 + 2$$

$$= 4 \text{ m}$$

M-II :

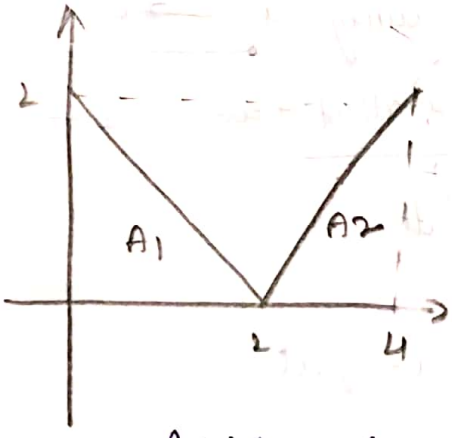


$A_1 = \frac{1}{2} \times 2 \times (-2) = -2$

$A_2 = \frac{1}{2} \times 2 \times 2 = 2$

for displacement,

$A = A_1 + A_2 = -2 + 2 = 0$



$A_1 + A_2 = A$

$A = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 2 \times 2$

$\boxed{A = 4}$

(Or) For distance,

$A = |A_1| + |A_2|$

$= |-2| + |2|$

$= 2 + 2$

$\boxed{A = 4}$

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M-III:

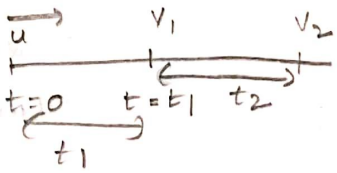
$$\text{Average velocity} = \frac{\int_0^4 v\,dt}{\int_0^4 dt} = \frac{0}{4} = 0 \text{ m/s}.$$

$$\text{Average speed} = \frac{\int_0^2 |v|\,dt + \int_2^4 |v|\,dt}{4} = \frac{4}{4} = 1 \text{ m/s}$$

* A particle has an acceleration a_1 for a time t_1 and an acceleration a_2 for a time t_2 then average acceleration is $\boxed{a = \frac{a_1 t_1 + a_2 t_2}{t_1 + t_2}}$.

proof:-

$$\langle a \rangle = \frac{\Delta v}{\Delta t} = \frac{v_2 - u}{t_1 + t_2}$$



$$a = \frac{v_2 - u}{t_1 + t_2}$$

$$v_1 = u + a_1 t_1$$

$$v_2 = v_1 + a_2 t_2$$

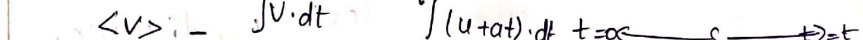
$$v_2 = u + a_1 t_1 + a_2 t_2$$

finally $a = \frac{a_1 t_1 + a_2 t_2}{t_1 + t_2}$

$a(t_1 + t_2)$ $\boxed{v_2 - u = a_1 t_1 + a_2 t_2}$

* Find average velocity of journey. $u \xrightarrow{a} v = u + at$

$$\langle v \rangle = \frac{\int v\,dt}{\int dt} = \frac{\int_0^t (u + at)\,dt}{\int_0^t dt}$$



$$\langle v \rangle = \frac{ut + \frac{1}{2}at^2}{t} = u + \frac{1}{2}at$$

$$\langle v \rangle = \frac{u + u + at}{2} = \frac{u + v}{2}.$$

Ivat
Sdt

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→ $v(t) = (t^2 - 2t)$ m/s. Find ① displacement for $0 \le t \le 3$, ② distance for $0 \le t \le 3$

$$v = t(t-2)$$

* $s \xrightarrow[\text{integrate}]{\text{diff}} v \xrightarrow[\text{integrate}]{\text{diff}} a$

$$v = \frac{ds}{dt}\ ;\ a = \frac{dv}{dt}\ ;\ a = v \cdot \frac{ds}{dt}$$

$$\int ds = \int v\,dt \ \&\ \int dv = \int a\,dt.$$

→ If the object starts from rest at $t = 0$ and from origin and accelerates at a rate given by $\boxed{a = 6t}$.

$$a = 6t$$

$$\frac{dv}{dt} = 6t$$

$$\int_0^v dv = \int_0^t 6t\,dt$$

$$v = \frac{3}{6/2}[t^2]_0^t$$

$$v = 3t^2 \Rightarrow t = \left(\frac{v}{3}\right)^{1/2}$$

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$v = 3t^2 \Rightarrow \frac{ds}{dt} = 3t^2$

$$\int_0^s ds = 3\int_0^t t^2 \cdot dt$$

$$s = \frac{3}{3}[t^3]_0^t$$

$$\boxed{s = t^3} \Rightarrow \boxed{t = s^{1/3}}$$

1

$$\left(\frac{v}{3}\right)^{1/2} = s^{1/3}$$

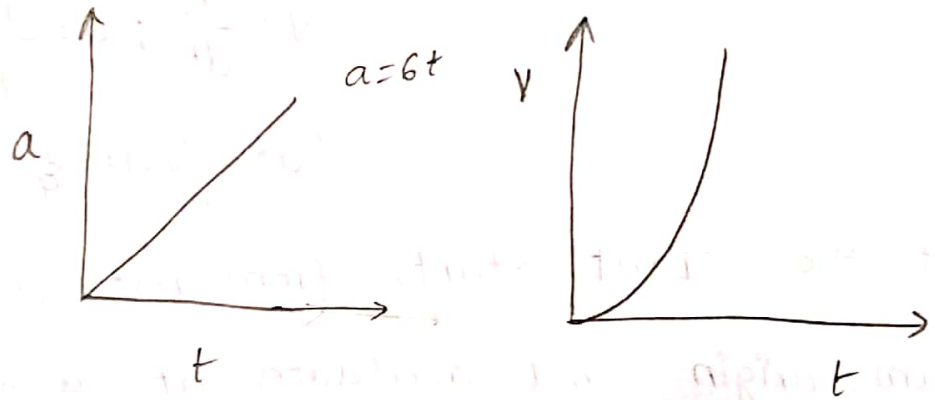
$$\frac{v}{3} = s^{2/3}$$

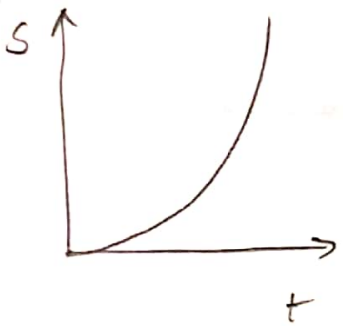
$$\boxed{v = 3s^{2/3}}$$

$$a = 6t = 6\left(\frac{v}{3}\right)^{1/2} = 2\sqrt{3v}$$

$$\boxed{a = 6s^{1/3}} \qquad \boxed{a = 2\sqrt{3v}}$$

graphs:-





→ For a particle moving along X-axis, acceleration is given by $\boxed{a = x}$. Find the position as a function of time. If at $t = 0$, $x = 1m$ and $v = 1m/s$.

Given $a = x$ $\qquad v \cdot \frac{dv}{dx} = x$

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$$\int_1^v v \cdot dv = \int_1^x x \cdot dx$$

$$\left[\frac{v^2}{2}\right]_1^v = \left[\frac{x^2}{2}\right]_1^x$$

$$v^2 - 1 = x^2 - 1$$

$$v^2 = x^2$$

$$\Rightarrow \boxed{v = x} \Rightarrow v = f(x)$$

now, $\frac{dx}{dt} = x$

$$\int_1^x \frac{dx}{x} = \int_0^t dt$$

$$[\ln x]_1^x = [t]_0^t$$

$$\ln x - \ln 1 = t$$

$$\ln x = t \Rightarrow \boxed{x = e^t} \Rightarrow x = f(t)$$

now,

$$v = \frac{dx}{dt} = \frac{d}{dt} e^t = e^t$$

$$\boxed{v = e^t} \Rightarrow v = f(t)$$

$$a = \frac{dv}{dt} = \frac{d}{dt} e^t$$

$$\boxed{a = e^t} \Rightarrow a = f(t)$$

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→ For a particle moving along x-axis, acceleration is given by $\boxed{a = 2v^2}$, if the speed of the particle is v_0 at $t=0$; Find the speed as a function of x. (and from origin)

$$a = 2v^2$$

$$\frac{dv}{dt} = 2v^2$$

$$\int_{v_0}^{v} \frac{dv}{2v^2} = \int_0^t dt$$

$$\frac{1}{2}\int_{v_0}^{v} v^{-2}\,dv = t$$

$$\left[\frac{v^{-1}}{-1}\right]_{v_0}^{v} = 2t$$

$$\frac{1}{v} - \frac{1}{v_0} = -2t$$

$$\frac{1}{v_0} - \frac{1}{v} = 2t$$

$$\frac{v - v_0}{v v_0} = 2t$$

$$\frac{1}{v_0} - \frac{1}{v} = 2t$$

$$\frac{1 - 2tv_0}{v_0} = \frac{1}{v} \Rightarrow \boxed{v = \frac{v_0}{1 - 2tv_0}} \Rightarrow v = f(t)$$

now, $$\frac{dx}{dt} = \frac{v_0}{1 - 2tv_0}$$

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$$\int_0^x dx = V_0 \int_0^t \frac{1}{1-2V_0 t} \cdot dt$$

$$x = \frac{V_0 \left[\ln(1-2V_0 t)\right]_0^t}{-2V_0}$$

$$x = -\frac{1}{2}\left[\ln(1-2V_0 t) - \ln(1)\right]$$

$$\boxed{x = -\frac{\ln(1-2V_0 t)}{2}} \Rightarrow x = f(t)$$

$$a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{V_0}{1-2V_0 t}\right) = V_0 \frac{(-1)(-2V_0)}{(1-2V_0 t)^2}$$

$$\boxed{a = \frac{2V_0^2}{(1-2V_0 t)^2}} \Rightarrow a = f(t)$$

and $a = 2v^2$

$$v \cdot \frac{dv}{ds} = 2v^2$$

$$\int_{V_0}^{V} \frac{dv}{2v} = \int_0^x ds$$

$$[\ln v]_{V_0}^{V} = 2x$$

$$\ln(v/V_0) = 2x$$

$$\frac{v}{V_0} = e^{2x}$$

$$\boxed{v = V_0 e^{2x}}$$

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→ motion of the body is given by $\frac{d}{dt}(v(t)) = 6 - 3v(t)$

$v(t)$ → speed in m/s, t → sec, body was at rest at $t = 0$ sec & $x = 0$ m.

1) Terminal Velocity :-

$$a = 0$$

$$6 - 3v = 0$$

$$6 = 3v$$

$$\boxed{v = 2 \text{ m/s}}$$

2) Initial acceleration :-

i.e, at $t = 0$, $v = 0$

$$\boxed{a_i = 6 \text{ m/s}^2}$$

3) $$\frac{dv}{dt} = 6 - 3v$$

$$\int_0^v \frac{dv}{6 - 3v} = \int_0^t dt$$

$$\frac{[\ln(6 - 3v)]_0^v}{-3} = t$$

$$\ln(6 - 3v) - \ln 6 = -3t$$

$$\ln\left(\frac{6 - 3v}{6}\right) = -3t$$

$$1 - \frac{v}{2} = e^{-3t}$$

$$\boxed{2(1 - e^{-3t}) = v} \Rightarrow v = f(t)$$

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now, $a = \frac{dv}{dt} = \frac{d}{dt}\left[2(1-e^{-3t})\right]$

$= \frac{d}{dt}2 - 2\frac{d}{dt}e^{-3t}$

$= -2(-3)\times e^{-3t} = 6e^{-3t}$

$\boxed{a = 6e^{-3t}}$

again,

$\frac{dx}{dt} = 2 - 2e^{-3t}$

$\int_0^x dx = \int_0^t (2 - 2e^{-3t}) \cdot dt$

$x = \int_0^t 2 \cdot dt - 2\int_0^t e^{-3t} \cdot dt$

$x = 2t + \frac{2}{+3}\left[e^{-3t}\right]_0^t$

$x = 2t + \frac{2}{3}\left[e^{-3t} - 1\right]$

$\boxed{x = 2t - \frac{2}{3}(1 - e^{-3t})}$

→ Velocity $V = \alpha\sqrt{x}$; α = const. At the moment $t = 0$, $x = 0 \Rightarrow \boxed{V = 0}$. Find

① $V = f(t)$ ② $a = f(t)$ ③ $S = f(t)$

④ Avg. velocity of the particle averaged over the time that the particle takes to cover first S-m of the point.

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$$v = \alpha\sqrt{x}$$

$$\frac{dx}{dt} = \alpha\sqrt{x}$$

$$\int_0^x \frac{dx}{\sqrt{x}} = \int_0^t \alpha \cdot dt$$

$$\int_0^x x^{-1/2} \cdot dx = \alpha t$$

$$\frac{x^{1/2}}{1/2} = \alpha t$$

$$2\sqrt{x} = \alpha t$$

$$\boxed{x = \frac{1}{4}\alpha^2 t^2} \Rightarrow x = f(t) \quad \text{— (3)}$$

now,

$$v = \frac{dx}{dt} = \frac{d}{dt}\left(\frac{1}{4}\alpha^2 t^2\right)$$

$$v = \frac{1}{4}\alpha^2 (2t)$$

$$\boxed{v = \frac{1}{2}\alpha^2 t} \Rightarrow v = f(t) \quad \text{— (1)}$$

&

$$a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{1}{2}\alpha^2 t\right)$$

$$\boxed{a = \frac{\alpha^2}{2}} = \text{const.} \quad \text{— (2)}$$

(4)

$$\langle v \rangle = \frac{\int_0^t v \cdot dt}{\int_0^t dt} = \frac{\int_0^t \frac{1}{2}\alpha^2 t \cdot dt}{t}$$

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$$= \frac{\left[\frac{1}{2}\alpha^2 \frac{t^2}{2}\right]_0^t}{[t]_0^t} = \frac{\frac{1}{4}\alpha^2 t^2}{t}$$

$$\langle v \rangle = \frac{1}{4}\alpha^2 t = \frac{1}{4}\alpha^2 \left(\frac{4x}{\alpha^2}\right)^{1/2}$$

$$\langle v \rangle = \frac{1}{2}\alpha^2 \times \frac{2\sqrt{x}}{\alpha}$$

$$\boxed{\langle v \rangle = \frac{1}{2}\alpha\sqrt{s}}$$

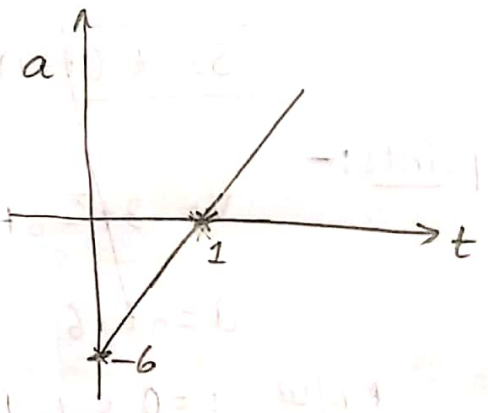
→ A particle moves along X-axis with acceleration $\boxed{a = 6(t-1)}$ where $t \to$ seconds. If the particle is initially at the origin and it moves along +ve X-axis with $v_0 = 2\,m/s$. Find the nature of motion of particle.

$a = 6(t-1) \Rightarrow a = 6t - 6$

$0 < t < 1$; $a = -ve$

$t > 1$; $a = +ve$

$t = 1$; $a = 0$



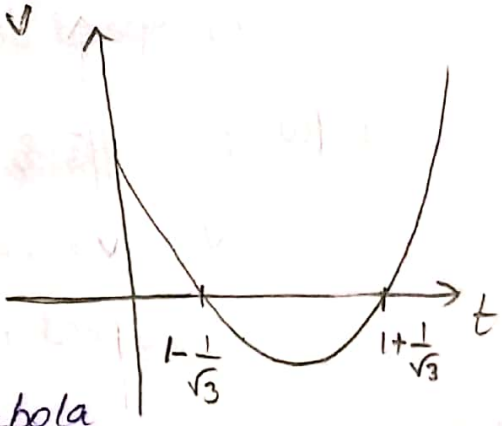
$$\frac{dv}{dt} = 6t - 6$$

$$\int_2^v dv = \int_0^t (6t-6)\,dt$$

$$v - 2 = \frac{6t^2}{2} - 6t$$

$$\boxed{v = 3t^2 - 6t + 2}$$

→ parabola



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$v = 0$, $\quad 3t^2 - 6t + 2 = 0$

$$t = \frac{6 \pm \sqrt{36 - 24}}{6} = \frac{6 \pm \sqrt{12}}{6} = \frac{6 \pm 2\sqrt{3}}{6}$$

$$\boxed{t = 1 \pm \frac{1}{\sqrt{3}}}$$

now,

$$\frac{ds}{dt} = 3t^2 - 6t + 2$$

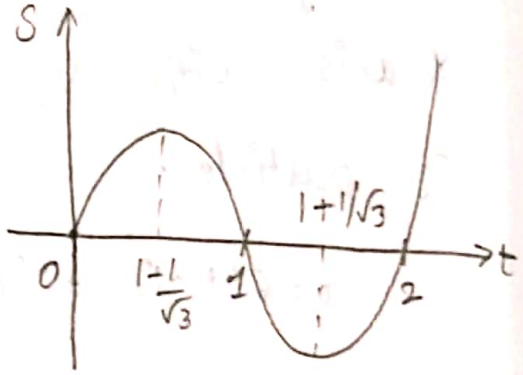
$$\int_0^s ds = \int_0^t (3t^2 - 6t + 2) \cdot dt$$

$$s = \frac{3t^3}{3} - \frac{6t^2}{2} + 2t$$

$$s = t^3 - 3t^2 + 2t$$

$$s = t(t^2 - 3t + 2)$$

$$\boxed{s = t(t-1)(t-2)}$$



Points :-

$$v = 3t^2 - 6t + 2$$

$$a = 6t - 6$$

* b/w $t = 0$ & $1 - 1/\sqrt{3}$ sec,

$v = +ve$, $a = -ve$

i.e, speed decreasing.

* b/w $t = 1 - 1/\sqrt{3}$ & 1 sec,

$v = -ve$, $a = -ve$.

i.e, speed increasing.

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* b/w $t = 1\text{sec}$ & $t = 1 + 1/\sqrt{3}\text{ sec}$;

$v = -ve$, $a = +ve$

i.e, speed decreasing.

* bfw $t > 1 + 1/\sqrt{3}$;

$v = +ve$, $a = +ve$

i.e, speed increases continuously.

* Maximum X-coordinate means $V = 0$

* Minimum [illegible] $= 0$

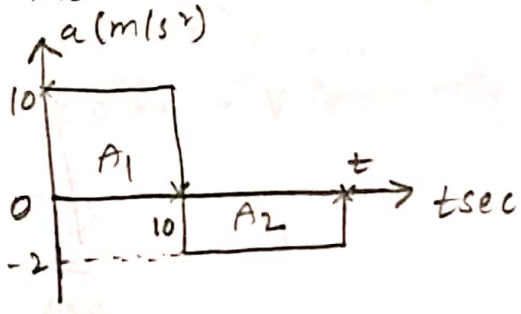
* V, a → opp direction → speed ↓

* V, a → same direction → speed ↑

→ An experimental car starts from rest & travels along a st. track such that it accelerates ~~along~~ at a const rate for 10 sec and deaccelerates at a const rate. draw v-t, s-t graphs & determine the time (t) needed to stop the car.

$$a = \frac{dv}{dt}$$

$$\int_{v_1}^{v_2} dv = \int_0^t a \cdot dt$$



$v_2 - v_1$ = Area of a-t graph

$A_1 = 10 \times 10$

$A_1 = 100$

$A_2 = (t' - 10)(-2)$

$A = A_1 + A_2 = 0$

$100 = 2(t' - 10)$

50

$t' = 60$

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from graph,

$a_1 = 10$ for $0 \le t < 10$

$a_2 = -2$ for $10 < t \le 60$

now, $\frac{dv_1}{dt} = 10$

$dv_1 = 10\,dt$

$$\boxed{v_1 = 10t} \quad -(1)$$

$$\frac{ds_1}{dt} = 10t$$

$$\int_0^{s_1} ds_1 = 10 \int_0^t t\,dt$$

$$s_1 = \frac{10\,t^2}{2}$$

$$\boxed{s_1 = 5t^2} \quad -(2)$$

* at $t = 10$ sec,

$v_1 = 10 \times 10 = 100$ m/s

& $s_1 = 5 \times 10^2$

$$\boxed{s_1 = 100\ m}$$

* for $10 < t \le 60$,

$a_2 = -2$

$$\frac{dv}{dt} = -2$$

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$$\int_{100}^{V_2} dV = -2 \int_{10}^{t} dt$$

$$V_2 - 100 = -2[t]_{10}^{t}$$

$$V_2 - 100 = -2t + 20$$

$$\boxed{V_2 = 120 - 2t}$$ for $10 \le t \le 60$

and $$\frac{ds}{dt} = 120 - 2t$$

$$ds = (120 - 2t) \cdot dt$$

$$\int_{500}^{S_2} ds = \int_{10}^{t} (120 - 2t) \cdot dt$$

$$S_2 - 500 = [120t - t^2]_{10}^{t}$$

$$S_2 - 500 = (120t - t^2) - (1200 - 100)$$

$$S_2 = -1100 + 500 + 120t - t^2$$

$$\boxed{S_2 = -600 + 120t - t^2}$$ position from origin

$$S_2 = -600 + 120 \times 20 - (20)^2$$

$$= -600 + 2400 - 400$$

$$\boxed{S_2 = 1400 \text{ m}}$$

Here dist, disp Equal

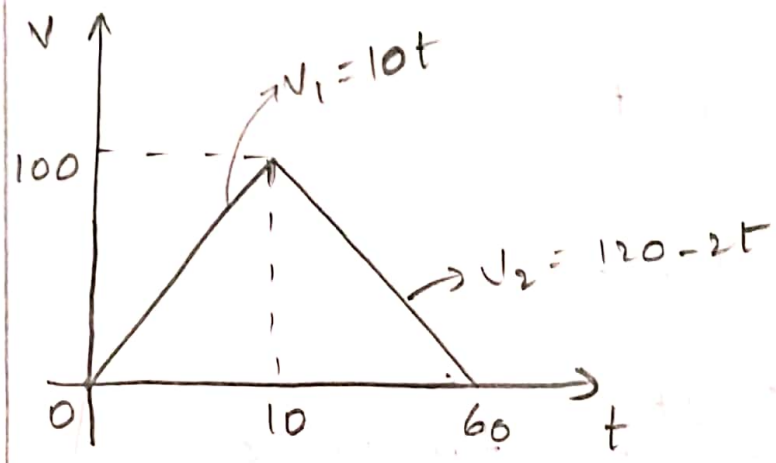
* distance travelled by body before coming to rest, $S = -600 + 120 \times 60 - (60)^2$

$$= -600 + 7200 - 3600$$

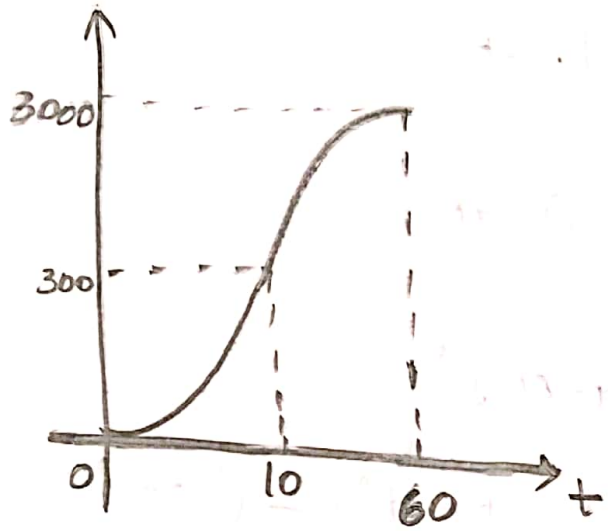
$$\boxed{S = 3000 \text{ m}}$$

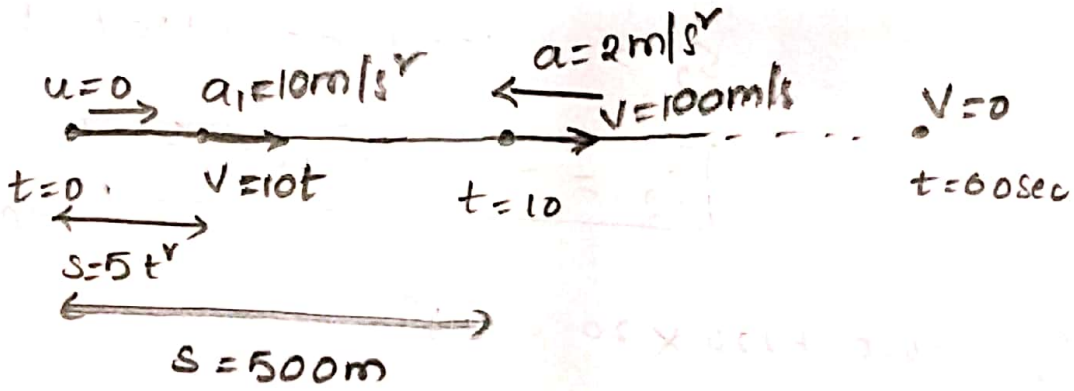
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V-t graph:



S-t graph:





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Kinematics (continuation)

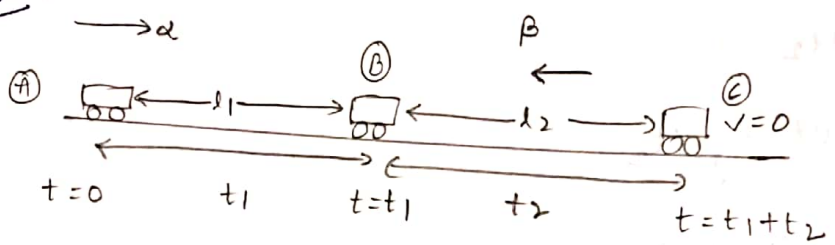
→ A car accelerates from rest at a constant rate 'α' for some time after which it deaccelerates at a constant rate β to come to rest. If the total time elapsed is t-secs. Find

① The maximum velocity reached

② Total distance travelled

③ Average velocity of journey.

sol: Method-I:



For AB:-

$$V = 0 + \alpha t_1 \Rightarrow \boxed{\alpha = V/t_1} \quad -①$$

$$l_1 = 0(t_1) + \frac{1}{2}\alpha t_1^2$$

$$\boxed{l_1 = \frac{1}{2}\alpha t_1^2} \qquad l_1 = \frac{1}{2}\left(\frac{V}{t_1}\right)t_1^2$$

$$\boxed{l_1 = Vt_1/2}$$

For BC :-

$$V_C = V_B + at$$

$$0 = V - \beta t_2$$

$$\boxed{\beta = V/t_2} \quad -②$$

and $$l_2 = Vt_2 + \frac{1}{2}(-\beta)t_2^2$$

$$l_2 = \beta t_2^2 - \frac{1}{2}\beta t_2^2$$

$$\boxed{l_2 = \frac{1}{2}\beta t_2^2} \qquad l_2 = \frac{1}{2} \times \left(\frac{V}{t_2}\right) \times t_2^2$$

$$\boxed{l_2 = \frac{1}{2}Vt_2}$$

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* If $\alpha = \beta$ then $\boxed{t_1 = t_2}$ & $\boxed{d_1 = d_2}$ like time of ascend = time of descend.

1) $t = t_1 + t_2$

$$t = \frac{V}{\alpha} + \frac{V}{\beta} = V\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$$

$$t = V\frac{(\alpha+\beta)}{\alpha\beta} \Rightarrow \boxed{V = \frac{\alpha\beta t}{\alpha+\beta}}$$

2) $d = d_1 + d_2$

$$= \frac{1}{2}Vt_1 + \frac{1}{2}Vt_2$$

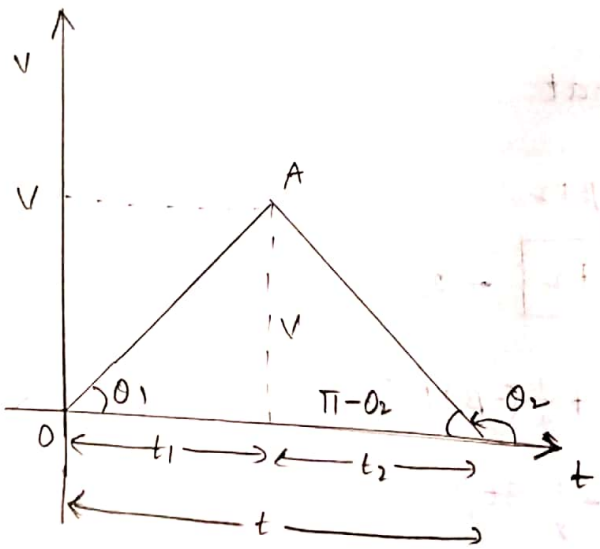
$$= \frac{1}{2}V(t_1 + t_2)$$

$$= \frac{1}{2}\frac{\alpha\beta t}{\alpha+\beta}(t)$$

$$\boxed{d = \frac{1}{2}\frac{\alpha\beta t^2}{(\alpha+\beta)}}$$

3) $\langle v \rangle = \frac{d}{t} = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right)t$ → average speed.

Method-II: Easy



$\tan\theta_1 = \alpha$ → acc

$\tan\theta_2 = -\beta$ = acc

From graph

$\tan\theta_1 = \frac{V}{t_1} = \alpha$

& $\tan(\pi - \theta_2) = -\tan\theta_2 = \frac{V}{t_2}$

$= +\beta = \frac{V}{t_2}$ $\Rightarrow \boxed{\beta = V/t_2}$

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now,

$t_1 + t_2 = t$

$t = \frac{V}{\alpha} + \frac{V}{\beta}$

$t = V\left(\frac{\alpha+\beta}{\alpha\beta}\right)$

$$\boxed{V = \frac{\alpha\beta t}{\alpha+\beta}}$$

l = Area of v-t graph

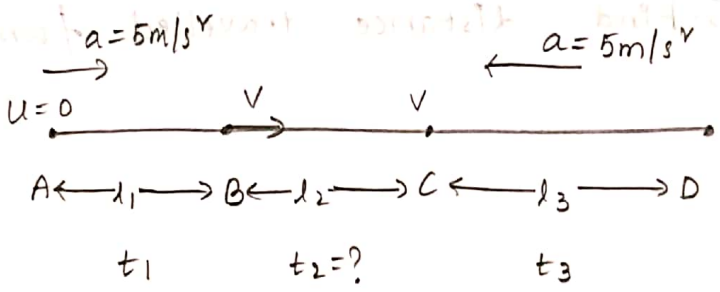
$l = \frac{1}{2} \times t \times V$

$$\boxed{l = \frac{1}{2}\frac{\alpha\beta t^2}{\alpha+\beta}}$$

$$\langle V \rangle = \frac{d}{t} = \frac{1}{2}\frac{\alpha\beta t}{\alpha+\beta}$$

→ A car starts moving rectilinearly first with an acceleration of $5\ m/s^2$ (initial velocity = 0) then moves uniformly and finally decelerating at the same rate till it stops the total time of Journey is 25 seconds. The Average speed during the interval is 72 Km/hr. Calculate the time for which the car was in uniform motion.

Sol:



$t_1 + t_3 + t_2 = 25$

$2t_1 + t_2 = 25 \rightarrow (1)$

$l_1 = \frac{1}{2} 5 t_1^2$

$l_2 = V t_2$

$l_2 = 5 t_1 t_2$

Avg speed $= \frac{l}{t} = \frac{l_1 + l_2 + l_3}{25} = 20$

$\Rightarrow l_1 + l_2 + l_3 = 500$

$2l_1 + l_2 = 500$

$2\left(\frac{1}{2}5t_1^2\right) + 5t_1 t_2 = 500$

$5t_1^2 + 5t_1(25 - 2t_1) = 500$

$5t_1^2 + 125t_1 - 10t_1^2 = 500$ $\Rightarrow$ $-5t_1^2 + 125t_1 - 500 = 0$ $\Rightarrow$

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$125t_1 - 5t_1^2 = 500$

$5t_1^2 - 125t_1 + 500 = 0$

$t_1^2 - 25t_1 + 100 = 0$

$t_1^2 - 5t_1 - 20t_1 + 100 = 0$

$t_1(t_1 - 5) - 20(t_1 - 5) = 0$

$(t_1 - 20)(t_1 - 5) = 0$

$t_1 = 5, 20$

✓ ✗

$t_2 = 25 - 2t_1 = 25 - 10$

$\boxed{t_2 = 15 \text{ sec}}$

→ Given, $a = -\cos t$ at $t = 0$, $u = 0$, $x = 1$. find position at $t = \frac{\pi}{2}$. Find distance travelled from 0 to 2π.

$a = -\cos t$

$\frac{dv}{dt} = -\cos t$

$\int_0^v dv = \int_0^t \cos t \cdot dt$

$\boxed{v = -\sin t}$

& $\frac{dx}{dt} = -\sin t$

$\int_1^x dx = \int_0^t -\sin t \cdot dt$

$x - 1 = [\cos t]_0^t$

$x - 1 = \cos t - \cos 0$

$x - 1 = \cos t - 1$

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$\Rightarrow \boxed{x = \cos t}$

At $t = \pi$

$x = \cos\pi = -1$

and Again,

$$\int_1^x dx = -\int_{t=0}^{2\pi} \sin t \cdot dt$$

$$x - 1 = -[\cos t]_0^{2\pi}$$

$$x - 1 = \cos 2\pi - \cos 0°$$

$$x - 1 = 1 - 1$$

$x - 1 = 0$

$\downarrow$ $\rightarrow x_i$

x_f (displacement)

For distance travelled,

$V = 0$

$\Rightarrow \sin t = 0$

$\Rightarrow t = 0, \pi, 2\pi, - - - -$

$$\int_1^x dx = \left| -\int_0^{\pi} \sin t \cdot dt \right| + \left| -\int_{\pi}^{2\pi} \sin t \cdot dt \right|$$

$$\int_1^x dx = \left| (\cos t)_0^{\pi} \right| + \left| (\cos t)_{\pi}^{2\pi} \right|$$

$$= |\cos\pi - \cos 0°| + |\cos 2\pi - \cos\pi|$$

$$= |-1-1| + |1-(-1)| = 2 + 2 = 4$$

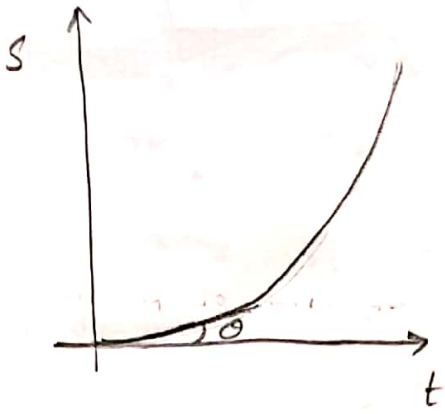
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$x - 1 = 4$

$\Downarrow$

(distance) //

Important points:-

* 

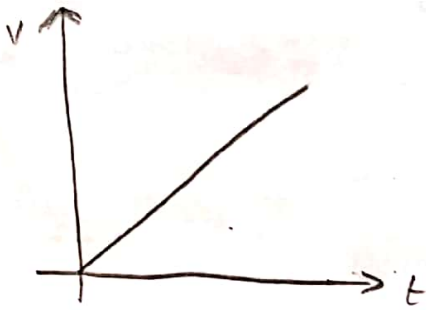
→ Slope = +ve & increasing

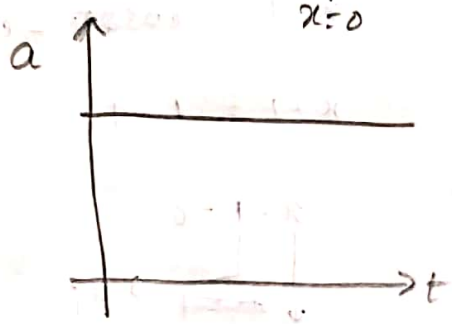
→ Velocity is increasing

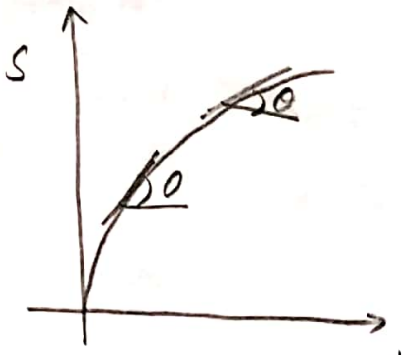
→ Speed in increasing

→ uniformly accelerated motion.







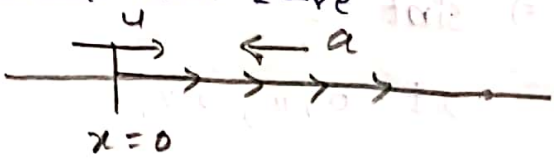
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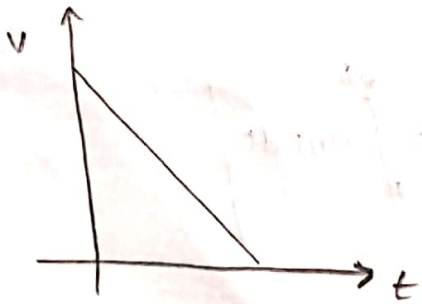
→ slope = +ve & decreasing

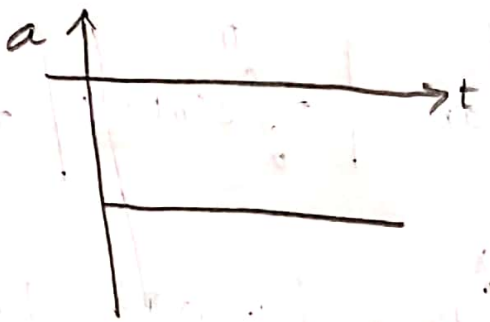
→ Velocity is decreasing

→ Speed is decreasing

→ acceleration = -ve







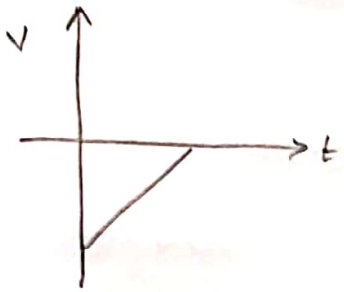
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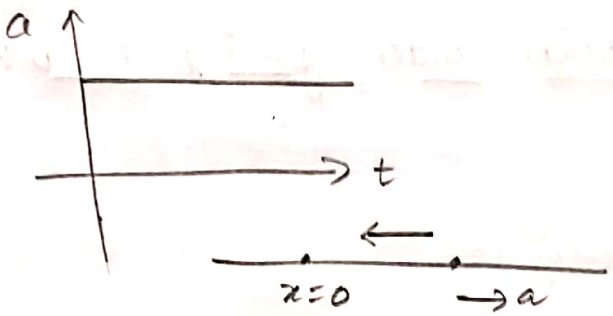
→ Slope = +ve ~~& decreas~~

→ Velocity is -ve & increasing

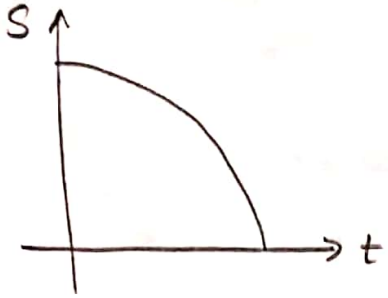
→ Speed is decreasing

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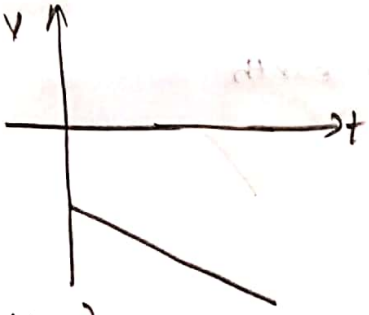


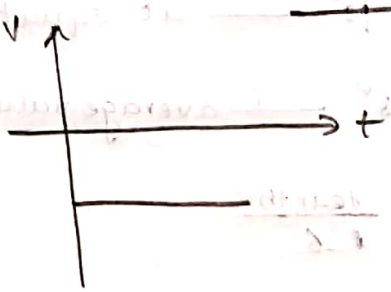
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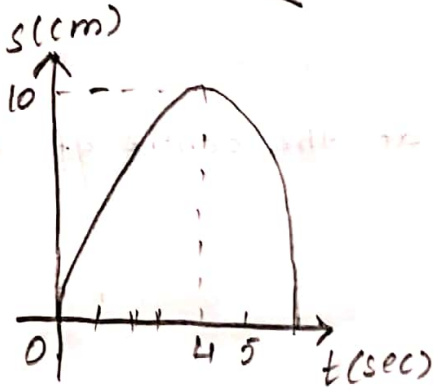
→ slope = -ve & decreasing
→ velocity = -ve
→ speed is increasing
→ accelerated motion.







*



$$S = \left(\frac{u+v}{2}\right)t$$

$$5 = \left(\frac{u+0}{2}\right)(4)$$

$$\boxed{u = 5\,cm/s}$$

& $v^2 - u^2 = 2as$

$$0^2 - 5^2 = 2a(10)$$

$$-25 = 20a$$

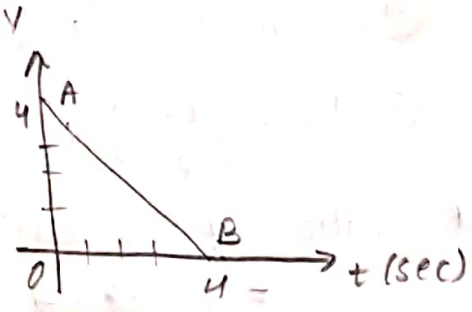
$$a = -5/4$$

$$\boxed{a = -1.25\,cm/s^2}$$

*

$$a = -\frac{4}{4} = -1\,cm/s^2$$

the graph will not show the same Motion.



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Motion under gravity in 1D :-

- Freely falling bodies
- Vertically projected bodies
- Combination

Acceleration due to gravity :

* $g = 9.83\ m/s^2$ — at poles

$g = 9.78\ m/s^2$ — at equator

$g = 9.8\ m/s^2$ — average value on earth

$g_{moon} = \frac{g_{earth}}{16}$

$g_{sun} = 274\ m/s^2$

* $g = 0$ at gravity free region & at the centre of earth of planet.

Equations of Motion :-

For dropped bodies — $u = 0$ $\quad a = g(\downarrow)$

(Air resistance is neglected)

(1) $v = u + at \Rightarrow v = gt$

(2) $s = ut + \frac{1}{2}at^2 \Rightarrow s = \frac{1}{2}gt^2$

(3) $v^2 - u^2 = 2as \Rightarrow v^2 = 2gs \Rightarrow v = \sqrt{2gs}$

(4) $S_n = u + a(n - \frac{1}{2}) \Rightarrow S_n = g(n - \frac{1}{2})$

(5) $s = \left(\frac{u+v}{2}\right)t$

* ratio of velocities at the end of successive seconds $= g : 2g : 3g : - - - -$

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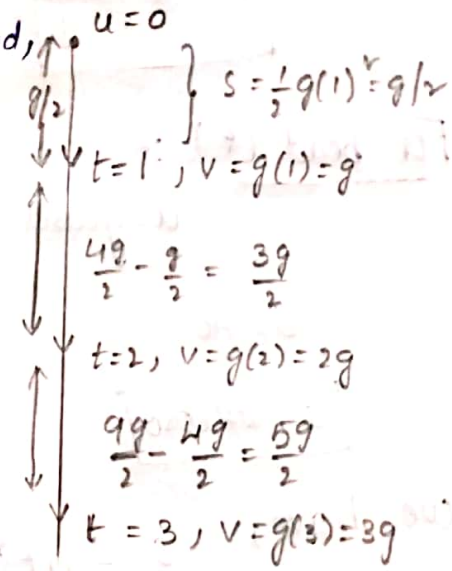
$= 1 : 2 : 3 : - - -$

* ratio of distance travelled in 1st second, first 2 seconds, 3 seconds - - - .

$= \frac{g}{2} : 4\left(\frac{g}{2}\right) : 9\left(\frac{g}{2}\right) : - - -$

$= 1 : 4 : 9 : - - -$

$= 1 : 2^2 : 3^2 : - - -$



* ratio of distances fallen during successive seconds

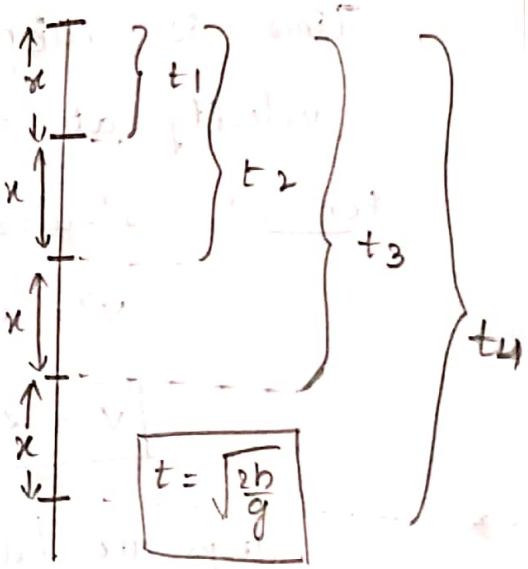
$= \frac{g}{2} : \frac{3g}{2} : \frac{5g}{2} : - - -$

$= 1 : 3 : 5 : - - - -$

* ratio of times taken to fall equal successive lengths.

$t_1 = \sqrt{\frac{2x}{g}}$, $t_2 = \sqrt{\frac{2(2x)}{g}}$,

$t_3 = \sqrt{\frac{2(3x)}{g}}$, $t_4 = \sqrt{\frac{2(4x)}{g}}$, - - -



$\therefore t_1 : t_2 - t_1 : t_3 - t_2 : t_4 - t_3 : - - -$

$= \sqrt{\frac{2x}{g}} : \sqrt{\frac{2x}{g}} - \sqrt{\frac{4x}{g}} : \sqrt{\frac{6x}{g}} - \sqrt{\frac{4x}{g}} : - - -$

$= \sqrt{1} : \sqrt{2} - \sqrt{1} : \sqrt{3} - \sqrt{2} : - - -$

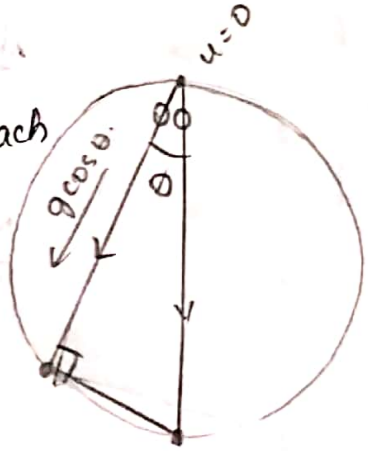
For bead (1) :- Find time of reach & velocity of beads.

$a = g$

$S = 2R$

$S = \frac{1}{2} g t^2 \Rightarrow 2R = \frac{1}{2} g t^2$





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$$\boxed{t = 2\sqrt{\frac{2R}{g}}}$$

For bead (2) :-

$$a = g\cos\theta$$

$$S = AC$$

$$\boxed{S = 2R\cos\theta}$$

$$\cos\theta = \frac{AC}{2R}$$

$$AC = 2R\cos\theta$$

we have, $S = \frac{1}{2}at^2$

$$2R\cos\theta = \frac{1}{2}g\cos\theta t^2$$

$$\boxed{t_2 = 2\sqrt{\frac{R}{g}}}$$

Time is independent of angle 'θ'.

velocity at the other end,

For 1 :- $v^2 - u^2 = 2as$

$$v^2 = 2g(2R)$$

$$\boxed{v = 2\sqrt{gR}}$$

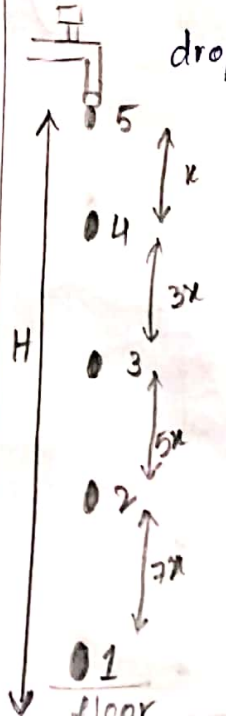
For 2 :- $v^2 - u^2 = 2as$

$$\boxed{v = 2\cos\theta\sqrt{gR}}$$

As $\theta\uparrow = \cos\theta\downarrow \Rightarrow v\downarrow$

→ drops are fallen in regular intervals.

Find distance b/w the drops.



M-I:

shortcut

$$x + 3x + 5x + 7x = H$$

$$16x = H$$

$$\boxed{x = H/16}$$

M-II:

time taken by the 1st drop to reach ground

$$t = \sqrt{\frac{2H}{g}}$$

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time gap b/w two successive drops

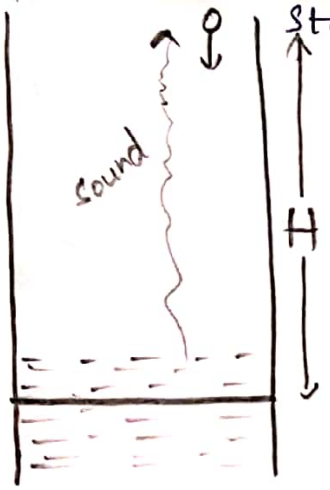
$$\Delta t = \frac{t}{n-1} = \frac{t}{5-1} = \frac{1}{4}\sqrt{\frac{2H}{g}}$$

* distance b/w 4th & 5th drop,

$$S_{45} = \frac{1}{2} g (\Delta t)^2$$

* distance b/w 2nd & 3rd drops,

$$S_{23} = \frac{1}{2} g (3\Delta t)^2 - \frac{1}{2} g (2\Delta t)^2$$



Stone is dropped from top of wall.

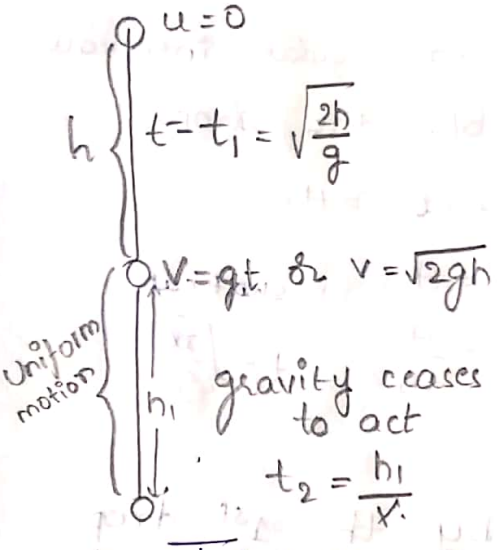
Time ~~(travelled)~~ taken by the sound to reach the person after it is dropped

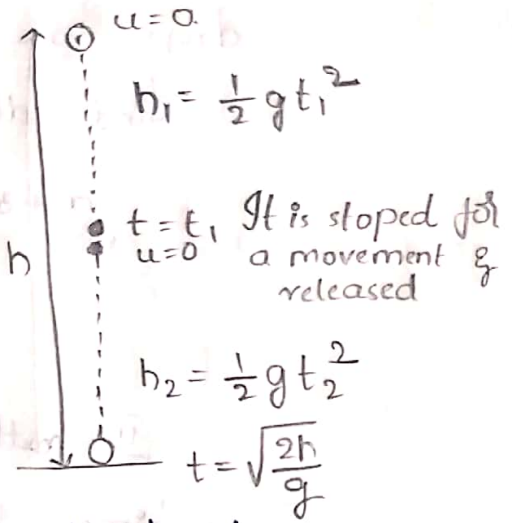
Here $t = t_1 + t_2$

$$= \sqrt{\frac{2H}{g}} + \frac{H}{V_{sound}}$$

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Ex 11





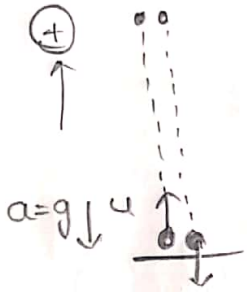
Total time $= t = t_1 + t_2$

$h = h_1 + h_2$

$\frac{1}{2}gt^2 = \frac{1}{2}gt_1^2 + \frac{1}{2}gt_2^2$

$\boxed{t^2 = t_1^2 + t_2^2}$

Vertically projected bodies:



Equations of motion:

1) $v = u + at \Rightarrow v = u - gt$

2) $s = ut + \frac{1}{2}at^2 \Rightarrow s = ut - \frac{1}{2}gt^2$

3) $v^2 - u^2 = 2as \Rightarrow v^2 = u^2 - 2gs$

$\Rightarrow \boxed{v = \pm\sqrt{u^2 - 2gs}}$

4) $S_n = u + a(n - \frac{1}{2}) \rightarrow S_n = u - g(n - \frac{1}{2})$

5) $s = \left(\frac{u+v}{2}\right)t$

→ 1) Maximum height:-

At $s = H$; $v = 0$

$v^2 - u^2 = 2as$

$0^2 - u^2 = 2(-g)(H)$

$\boxed{H = \frac{u^2}{2g}}$

Clearly $H \propto u^2$ [if g = const]

& $H \propto \frac{1}{g}$ [if u = const]

2) Time of ascent (t_a):

$v = u + at$

at $t = t_a$; $v = 0$

$0 = u - gt_a$

$\boxed{t_a = u/g}$

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3) Time of descent (t_d):

$S = ut + \frac{1}{2}at^2$

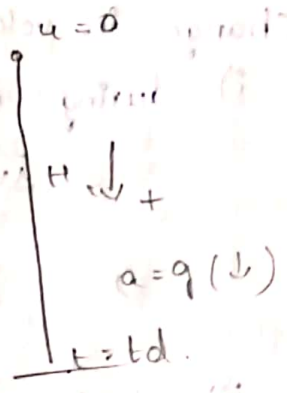
$H = 0(t_d) + \frac{1}{2}(g)(t_d)^2$

$H = \frac{1}{2}g\,t_d^2$

$\frac{u^2}{2g} = \frac{g}{2}(t_d)^2$

$t_d^2 = \frac{u^2}{g^2}$

$t_d = \frac{u}{g}$



* In the absence of air resistance $t_a = t_d$

4) Time of flight: (T):

$T = t_a + t_d$

$= \frac{u}{g} + \frac{u}{g} = \frac{2u}{g}$

Relation b/w H & T:

$H = \frac{u^2}{2g}$ & $T = \frac{2u}{g} \Rightarrow u = \frac{Tg}{2}$

$H = \frac{T^2 g^2}{2g \times 4} = \frac{1}{8}gT^2$

$$\boxed{H = \frac{1}{8}gT^2}$$

5) Velocity on reaching the point of projection:

$v^2 - u^2 = 2as$

For a complete journey

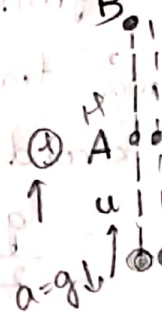
$v^2 - u^2 = 2(-g)(0)$

$v^2 - u^2 = 0$

$v^2 = u^2$

$v = \pm u$

clearly $v = -u$ & $v = u$



(In the absence of air resistance)

$v_A = v_C$

$t_{AB} = t_{BC}$

Here A & C coincidence.

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Points:

① Change in velocity

i) during upward journey

$$\Delta\vec{v} = \vec{v_f} - \vec{v_i} \quad (+)$$
$$= 0 - u$$
$$\vec{\Delta v} = -u$$
$$\Delta v = u, \text{ downward.}$$

ii) during downward journey

$$\Delta\vec{v} = \vec{v_f} - \vec{v_i} \quad (+)$$
$$= v - 0$$
$$\Delta v = v, \text{ downwards}$$

iii) For a complete journey

$$\Delta\vec{v} = \vec{v_f} - \vec{v_i} \quad (+)$$
$$\Delta\vec{v} = -v - u \quad (v = u)$$
$$= -2u$$
$$\Delta v = 2u, \text{ downwards}$$

② Average velocity: for a complete journey

$$\langle v \rangle = \frac{0}{T} = 0.$$

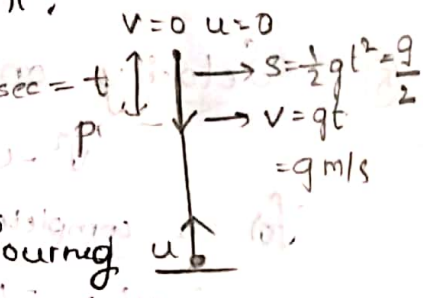
$$\text{Average speed} = \frac{2H}{T} = \frac{2}{2u/g} \times \frac{u^2}{2g} = \frac{u}{2},$$

irrespective of initial velocity.

③ Velocity of the body one second before reaching the maximum height;

Reverse

$$= g \text{ m/s} = 9.8 \text{ m/s}.$$



& distance travelled during last second of upward journey

$$= \frac{g}{2} = 4.9 \text{ m}.$$

④ distance travelled during first second of upward journey = distance travelled during last second of downward journey

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$S_n = u + a\left(n - \frac{1}{2}\right)$

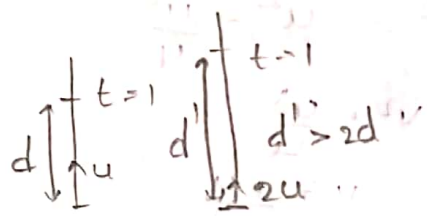
$= u - g\left(1 - \frac{1}{2}\right)$

$S_n = u - \frac{g}{2}$ = distance travelled during last & first second of downward & upward journey

It depends on "u"

For ex

If u is doubled then d'



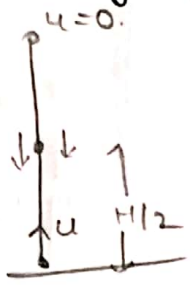
$d' = 2u - \frac{g}{2}$ $\quad d = u - \frac{g}{2}$ $\quad u = d + \frac{g}{2}$

$d' = 2d + g - \frac{g}{2}$

$d' = 2d + \frac{g}{2}$

$d' > 2d.$

ii) Velocity at half of the maximum height,



$v^2 - u^2 = 2as$

$v^2 - u^2 = 2(-g)\left(\frac{H}{2}\right)$

$v^2 - u^2 = -g\left(\frac{u^2}{2g}\right)$

$v^2 - u^2 = -\frac{u^2}{2}$

$v^2 = u^2 - \frac{u^2}{2}$ $\quad v^2 = \frac{u^2}{2}$

$v = \pm \frac{u}{\sqrt{2}}$

Time taken by the body to reach half of the maximum height

$v = u + at$

$\frac{u}{\sqrt{2}} = u - gt$ $\quad \Rightarrow gt = u - \frac{u}{\sqrt{2}}$

$gt = \frac{u}{g}\left(1 - \frac{1}{\sqrt{2}}\right)$

$t = ta\left(1 - \frac{1}{\sqrt{2}}\right)$

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(6) Velocity at $\frac{3}{4}$th of maximum height;

$v^2 - u^2 = 2as$

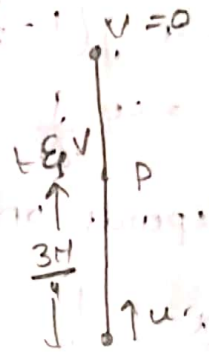
$v^2 - u^2 = 2(-g)\left(\frac{3H}{4}\right)$

$v^2 - u^2 = -\frac{3g}{2}\cdot\frac{u^2}{2g}$

$v^2 = u^2 - \frac{3u^2}{4}$

$v^2 = \frac{u^2}{4}$

$v = \pm\frac{u}{2}$



∴ time taken. $v = u + at$

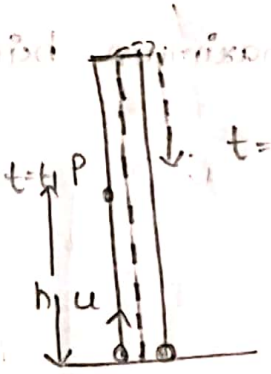
$\frac{u}{2} = u + at$

$at = -\frac{u}{2}$

$-gt = -\frac{u}{2}$

$\boxed{t = \frac{u}{2g} = \frac{ta}{2}}$

Ex:



t_1 & t_2 are the times taken from the instant of projection.

then,

1) Time of flight

$T = t_1 + t_2 = \frac{2u}{g}$

∴ Initial velocity of projection $u = \frac{g}{2}(t_1 + t_2)$

2) To find height of point P:

$s = ut + \frac{1}{2}at^2$

$h = -ut + \frac{1}{2}(-g)t^2$

$gt^2 = 2ut - 2h$

$gt^2 - 2ut + 2h = 0.$

↳ roots t_1 & t_2

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Sum of roots

$$t_1 + t_2 = \frac{2u}{g}$$

& $$t_1 t_2 = \frac{2h}{g}$$

$$h = \frac{g}{2} t_1 t_2$$

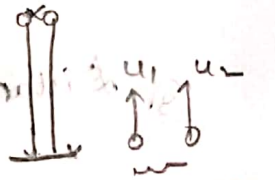
3) If the body drops from same height "h" then time taken to reach the ground is

$$h = \frac{1}{2} g t^2$$

$$\Rightarrow \frac{g}{2} t_1 t_2 = \frac{g}{2} t^2$$

$$t = \sqrt{t_1 t_2}$$

Two bodies are projected from the ground with speeds u_1 & u_2 with a time gap of n sec's. Find the time after which they meet ?



$$h = u_1 t - \frac{1}{2} g t^2 = u_2 (t-n) - \frac{1}{2} g (t-n)^2$$

$$u_1 t - \frac{1}{2} g t^2 = u_2 t - u_2 n - \frac{1}{2} g t^2 - \frac{1}{2} g n^2 + g t n$$

$$\frac{1}{2} g n^2 + u_2 n = (u_2 - u_1) t + g t n$$

$$t = \frac{u_2 n + \frac{g n^2}{2}}{(u_2 - u_1) + g n} \quad \text{for } 1^{st} \text{ body} \rightarrow (2)$$

For second body $t' = t - n$.

If $u_1 = u_2 = u$, then $t = \dfrac{un + \frac{gn^2}{2}}{gn}$

$$t = \frac{u}{g} + \frac{n}{2} \quad \text{for } 1^{st} \text{ body}$$

& for 2nd body,

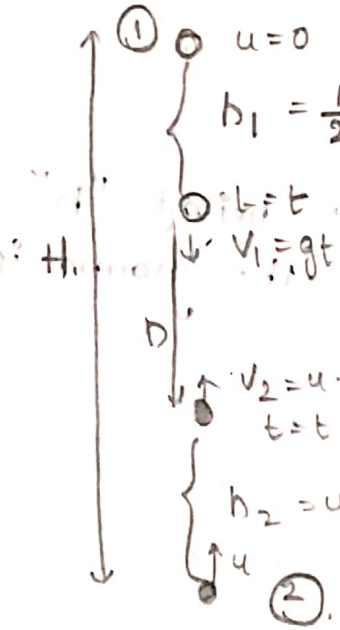
$$t' = t - n = \frac{u}{g} + \frac{n}{2} - n \Rightarrow t' = \frac{u}{g} - \frac{n}{2}$$

To find the height at which they collide sub eq (2) in eq (1)

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sub eq ② in eq ①

★★★ From top of building: (simultaneous action)



① Distance b/w two bodies after time t,

$h + h_1 + h_2 = H$

$h = H - (h_1 + h_2)$

$= H - \left(\frac{1}{2}gt^2 + ut - \frac{1}{2}gt^2\right)$

$\boxed{h = H - ut}$

② When they collide.

$h = 0$

$H - ut = 0$

$\boxed{t = \frac{H}{u}}$ ★

③ Distance travelled by both the bodies.

$h_1 + h_2 = H$

→ Velocities at the time of collision.

$V_1 = gt = g\left(\frac{H}{u}\right) = \frac{gH}{u}$

& $V_2 = u - gt = u - g\left(\frac{H}{u}\right)$

④ the time at which magnitude of their velocity are equal.

$V_1 = V_2$

$gt = u - gt$

$2gt = u$

$t = \frac{u}{2g}$

In this direction

$S_1 = \frac{1}{2}gt^2 = \frac{1}{2}g\left(\frac{u^2}{4g^2}\right) = \frac{u^2}{8g}$

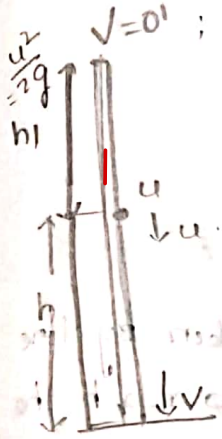
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& $S_2 = ut - \frac{1}{2}gt^2 = u\left(\frac{u}{2g}\right) - \frac{g}{2}\left(\frac{u}{2g}\right)^2$

$$S_2 = \frac{u^2}{2g} - \frac{u^2}{8g} = \frac{3u^2}{8g}$$

$$S_1 : S_2 = 1 : 3$$

Vertical projected body from certain height:



Time taken to reach the ground:

$$s = ut + \frac{1}{2}at^2$$

$$-h = ut + \frac{1}{2}(-g)t^2$$

$$gt^2 - 2ut - 2h = 0.$$

$$t = \frac{2u \pm \sqrt{(-2u)^2 + 8gh}}{2g}$$

$$t = \frac{2u \pm 2\sqrt{u^2+2gh}}{2g}$$

$$t = \frac{u \pm \sqrt{u^2+2gh}}{g}$$

$$t_1 = \frac{u}{g} - \frac{\sqrt{u^2+2gh}}{g} \times$$ less than ta

$$\boxed{t_2 = \frac{u}{g} + \frac{\sqrt{u^2+2gh}}{g}} \checkmark$$

* Velocity while striking ground:

$$v^2 - u^2 = 2as$$

$$(-v)^2 - u^2 = 2(-g)(-H)$$

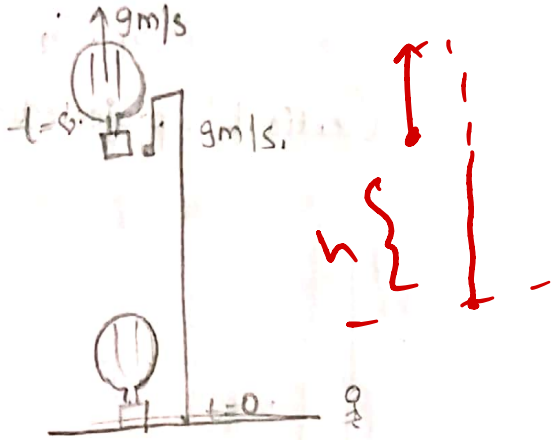
$$v^2 = u^2 + 2gH$$

$$\boxed{v = \sqrt{u^2+2gH}}$$

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Ex.

A balloon starts from rest moves vertically upwards with an acceleration $g/8$ m/s² a stone falls from the balloon after 8 sec from the fall find the time taken by the stone to reach the ground.



i) Velocity of ballon at the end of 8 secs

$v = u + at$

$v = at$

$v = \frac{g}{8} \times 8, \quad v = g$ m/s

⇒ height of balloon from ground after $t = 8$ sec

$h = \frac{1}{2}at^2 = \frac{1}{2}\frac{g}{8} \times 8 \times 8 = 4g$

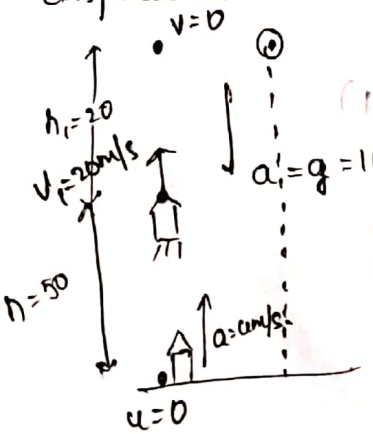
Time taken by the stone to reach the ground $4g = \frac{g}{2}t^2 - gt$

$h = \frac{1}{2}gt^2 - ut$

$4g = \frac{1}{2}gt^2 - ut$

$t = 4$ sec

A rocket is fired vertically upwards with a net acceleration of 4 m/s² & initial velocity zero after 5 sec its fuel is finished and decelerates with g at highest point, it's velocity becomes zero there after it accelerates downwards with acceleration g & return back to ground. Plot v-t graph for the total journey & also displacement - time graph. ($g = 10$ m/s²).



Step 1:

i) Velocity of rocket at the end 5 sec (fuel is empty)

$v = at = 4 \times 5 = 20$ m/s

→ height attained by rocket till 5 secs

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$h = \frac{1}{2}at^2 = \frac{1}{2} \times 4 \times 5 \times 5 = 50m$

Step 2:

time of ascent

$t = \frac{V_1}{g} = \frac{20}{10} = 2sec.$

& $h_1 = \frac{V_1^2}{2g} = \frac{20 \times 20}{2 \times 10} = 20m.$

Step 3:

time taken to reach the ground

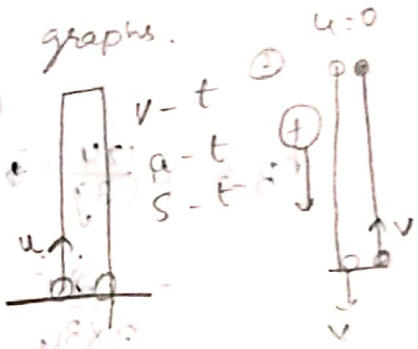
$(h + h_1) = \frac{1}{2} g t_1^2$

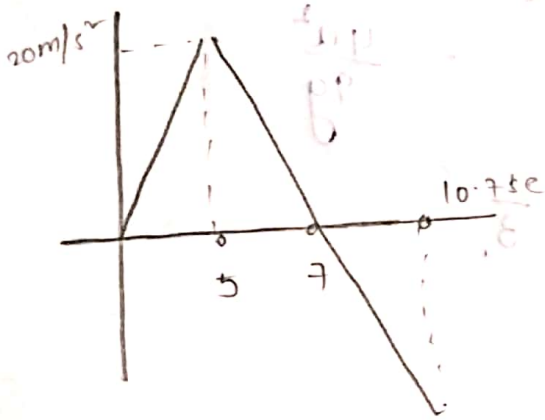
$70 = \frac{1}{2} \times 10 \times t_1^2$

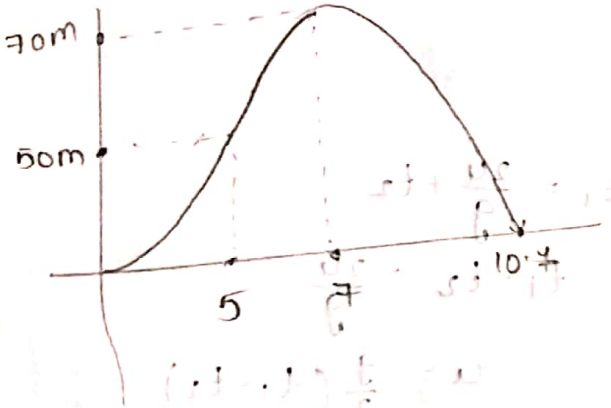
$t_1^2 = 14$

$t_1 = \sqrt{14} = 3.7 sec$

graphs.

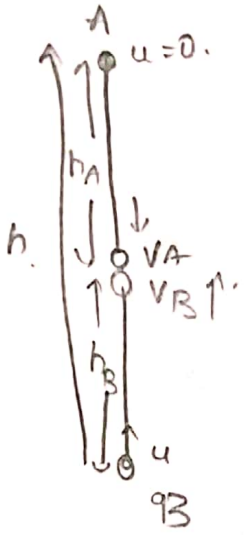






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If $V_A = 2V_B$. find the fraction of height at which bodies collide



If $V_A = 2V_B$

$gt = 2(u - gt)$

$gt + 2gt = 2u$

$3gt = 2u$

$t = \frac{2u}{3g}$

$h_A = \frac{1}{2} g t^2 = \frac{1}{2} g \frac{2u^2}{9g^2} = \frac{2u^2}{9g}$

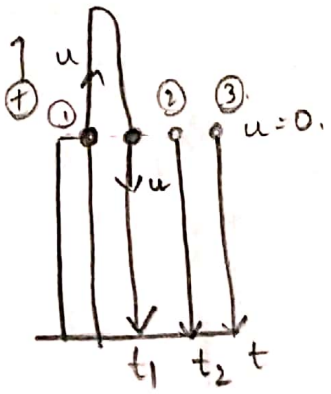
$h_B = u\left(\frac{2u}{3g}\right) - \frac{1}{2} g \left(\frac{4u^2}{9g^2}\right)$

$= \frac{6u^2}{3 \times 3g} - \frac{2u^2}{9g} = \frac{4u^2}{9g}$

$\frac{h_A}{h_A + h_B} = \frac{\frac{2u^2}{9g}}{\frac{6u^2}{9g}} = \frac{1}{3}.$

$\frac{h_B}{h_A + h_B} = \frac{\frac{4u^2}{9g}}{\frac{6u^2}{9g}} = \frac{2}{3}.$

ex:



$t_1 = \frac{2u}{g} + t_2$

$t_1 - t_2 = \frac{2u}{g}$

$u = \frac{g}{2}(t_1 - t_2)$

For 1st body

$h = -ut_1 + \frac{1}{2} g t_1^2 \rightarrow ①$

For 2nd body

$h = ut_2 + \frac{1}{2} g t_2^2 \rightarrow ①$

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now from eq ①

$$h = \frac{1}{2}gt_1^2 - \frac{g}{2}(t_1 - t_2)t_1$$

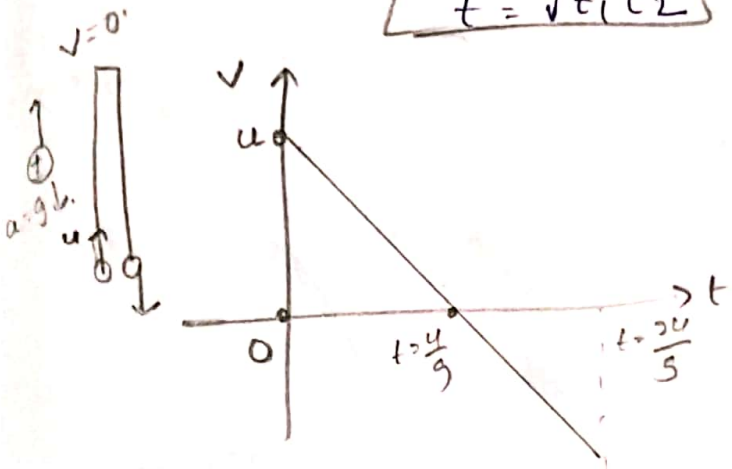
$$= \frac{1}{2}gt_1^2 - \frac{g}{2}t_1^2 + \frac{g}{2}t_1t_2$$

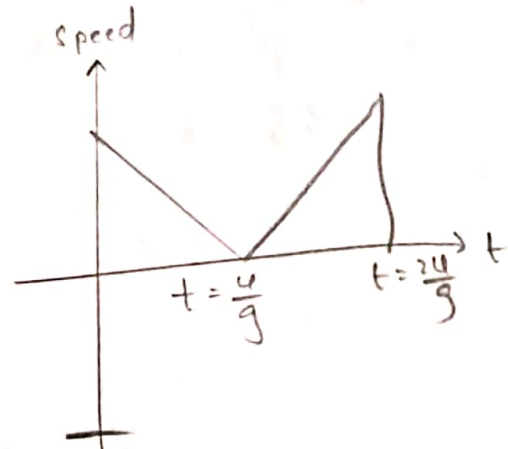
$$\boxed{h = \frac{1}{2}gt_1t_2}$$

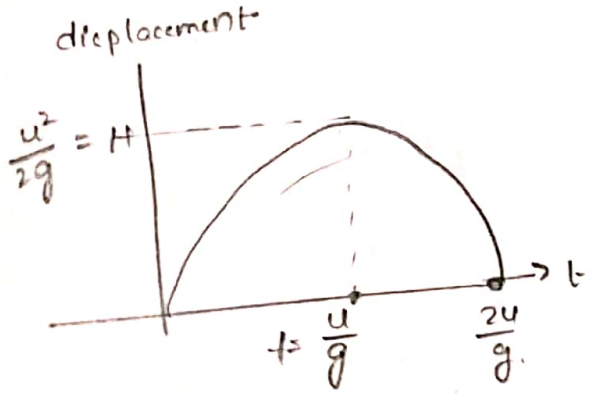
If 3rd body is dropped from same height, then $h = \frac{1}{2}gt^2$

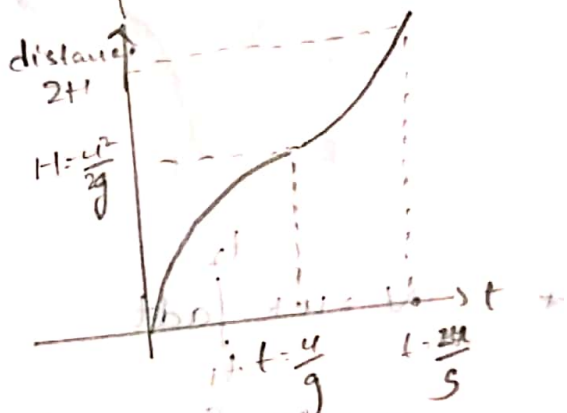
$$\frac{1}{2}gt_1t_2 = \frac{1}{2}gt^2$$

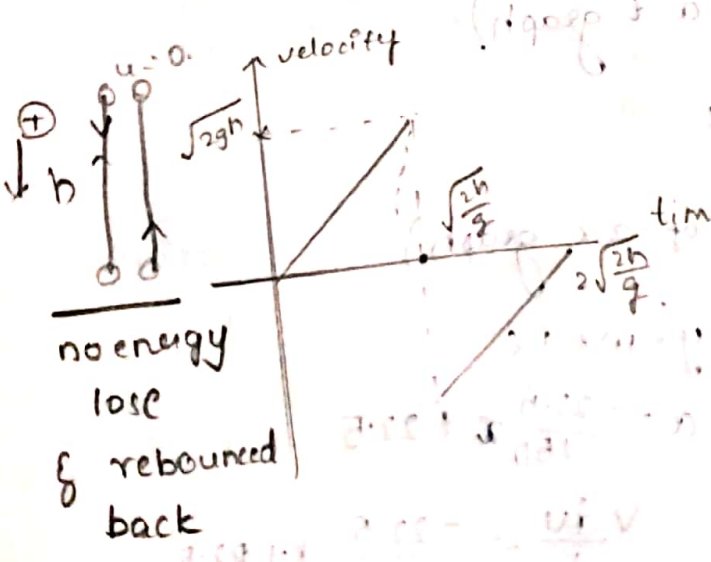
$$\boxed{t = \sqrt{t_1t_2}}$$

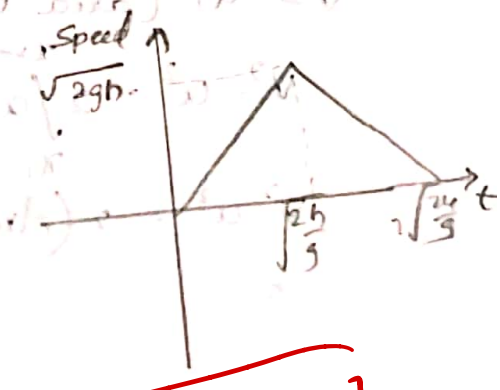






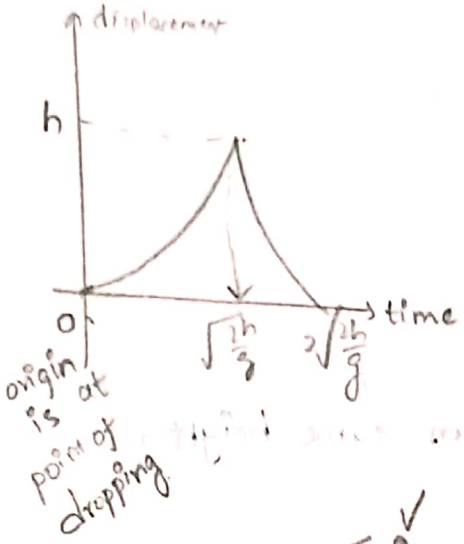


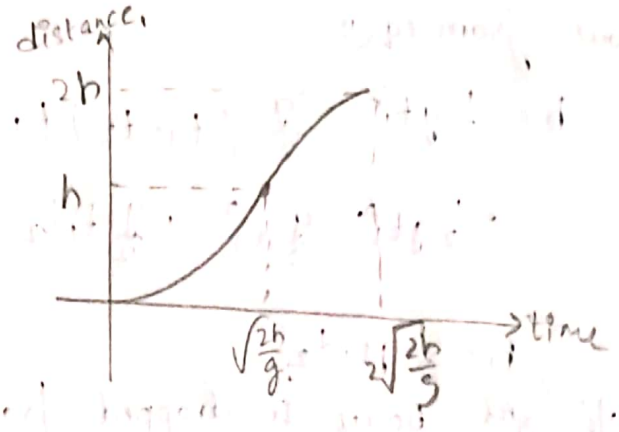


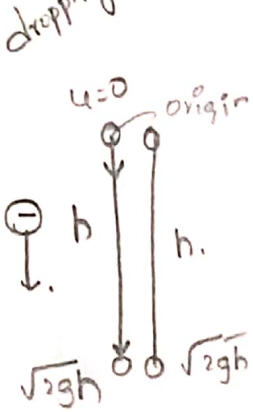


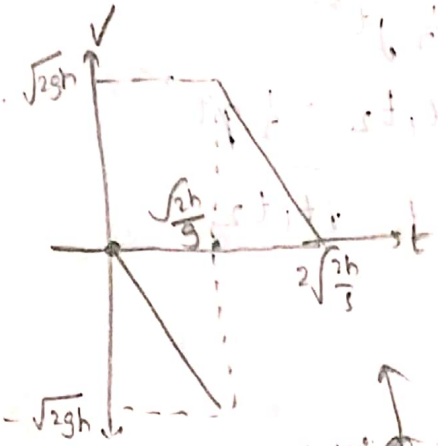
MR SIR

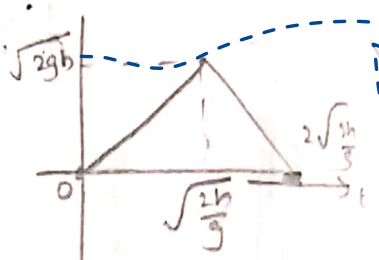
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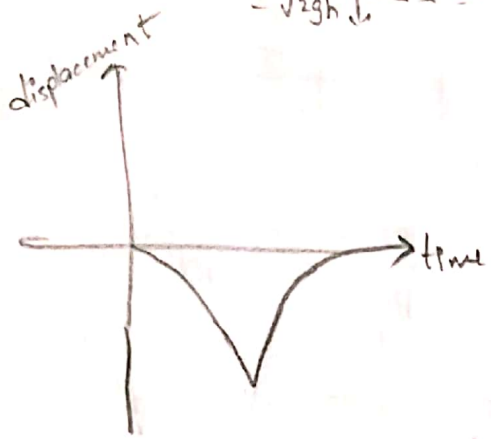


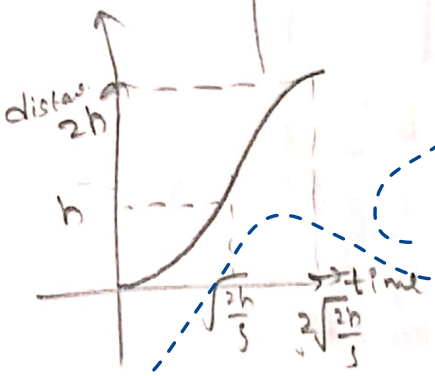










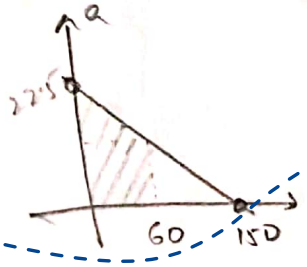


$$v = u + \int_{t_1}^{t_2} a\,dt$$

$v = u +$ (area of a-t graph)

$$v^2 - u^2 = 2\int_{x_1}^{x_2} a\,dx$$

$v^2 - u^2 = 2$ (Area of a-x graph)



$y = mx + c$

$a = \frac{-22.5}{150}x + 22.5$

$v\frac{dv}{dx} = \frac{-22.5}{150}x + 22.5$

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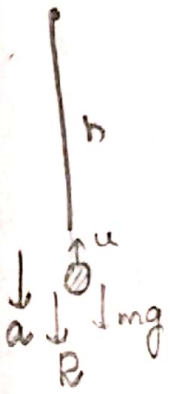
$$\int_u^v v\,dv = \int_{x_1}^{x_2} \frac{-22.5}{150}x + 22.5$$

(ii) method

At $x = 60$

$$a = \frac{-22.5}{150} \times 60 + 22.5$$

In the presence of constant air resistance:



$$F_{net} = mg + R$$

$$ma = mg + R$$

$$\text{deceleration} = g + \frac{R}{m}$$

clearly deceleration $> g$ ✓

now,

Time of ascent: $v = u + at$

$$0 = u - \left(g + \frac{R}{m}\right) t_a$$

$$\boxed{t_a = \frac{u}{g + \frac{R}{m}}}$$ → t_a is less than time of ascent in vaccum.

replace g by $g + \frac{R}{m}$

& $h = \frac{u^2}{2\left(g + \frac{R}{m}\right)}$

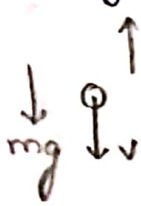
$$= \frac{\left(g + \frac{R}{m}\right)^2 t_a^2}{2\left(g + \frac{R}{m}\right)} \Rightarrow \boxed{h = \frac{1}{2}\left(g + \frac{R}{m}\right) t_a^2}$$

& $v^2 - u^2 = 2as$

$$0^2 - u^2 = -2\left(g + \frac{R}{m}\right) h$$

$$\boxed{u = \sqrt{2\left(g + \frac{R}{m}\right) h}}$$

During descent



net force

$$F_{net} = mg - R$$

$$ma = mg - R$$

$$a = \frac{g}{1} - \frac{R}{m}.$$

acceleration $< g$ ✓

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We have

$$h = \frac{1}{2}\left(g - \frac{R}{m}\right)t_d^2$$

now

$$\frac{1}{2}\left(g + \frac{R}{m}\right)t_a^2 = \frac{1}{2}\left(g - \frac{R}{m}\right)t_d^2$$

clearly $t_d > t_a$

now $T = t_a + t_d$; may increase, decrease & same

$$\frac{V}{u} = \frac{\sqrt{\frac{1}{2}\left(g - \frac{R}{m}\right)h}}{\sqrt{\frac{1}{2}\left(g + \frac{R}{m}\right)h}} < 1.$$

$$\frac{V}{u} < 1 \qquad V < u.$$

For droped bodies:

acceleration $a = g - \frac{R}{m}$

1) Resistance is proportional to mass $\frac{R}{m}$ is same for all bodies

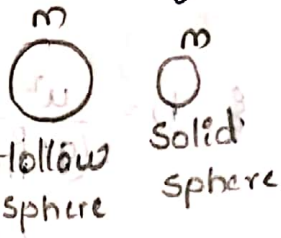
$\Rightarrow$ 'a' is same for all bodies

All bodies reach the ground simultaneously

2) If m is constant

$\therefore$ a is less for bigger sized bodies.

Smaller bodies will reach the ground earlier



3) If R is same for all bodies

$m \uparrow \quad \frac{R}{m} \downarrow \Rightarrow a \uparrow$

body with high mass come to ground earlier.

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